

ZID-Net: Zero-Inference Diffusion Prior Decoupling Network for Single Image Dehazing★

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Abstract

Single image dehazing is often constrained by a trade-off between restoration quality and computational efficiency. While efficient, CNNs struggle to learn robust priors for dense and non-homogeneous haze. Conversely, diffusion models provide strong generative priors but suffer from severe inference latency and sampling instability. To address these limitations, we propose ZID-Net, a novel framework that explicitly decouples diffusion supervision from feed-forward inference. For efficient inference, we design a frequency-spatial decoupled feed-forward backbone. Within this backbone, a Channel-Spatial Laplacian Mask (CSLM) filters haze-amplified noise to extract purified structural details, while Lightweight Global Context Blocks (LGCBs) establish long-range spatial dependencies to capture global haze variations. A Dynamic Feature Arbitration Block (DFAB) then adaptively fuses these semantic and structural features for robust reconstruction. To provide the backbone with degradation-aware priors without increasing inference cost, we introduce a Zero-Inference Prior Propagation Head (ZI-PPH) during training. ZI-PPH leverages a conditional diffusion process to predict residual noise and regularize

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The project code is available at: <https://github.com/XoomitLXH/ZID-Net>.
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ZID-Net: 零推理扩散先验 用于单图像去雾的解耦网络 ★

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摘要

单图像去雾常常受到复原质量与计算效率之间的权衡制约。卷积神经网络虽然高效，但难以学习适用于浓雾和非均匀雾霾的稳健先验。相反，扩散模型提供了强大的生成先验，但存在严重的推理延迟和采样不稳定性。为了解决这些局限性，我们提出了 ZID-Net，这是一种将扩散监督与前馈推理明确解耦的新型框架。为了实现高效推理，我们设计了一个频域-空间解耦前馈骨干网络。在该骨干网络中，通道-空间拉普拉斯掩码（CSLM）过滤雾放大噪声以提取纯净的结构细节，而轻量级全局上下文模块（LGCBs）建立长程空间依赖以捕获全局雾度变化。随后，动态特征仲裁模块（DFAB）自适应地融合这些语义与结构特征以实现稳健重建。为了在不增加推理成本的情况下为骨干网络提供退化感知先验，我们在训练期间引入了零推理先验传播头（ZI-PPH）。ZI-PPH 利用条件扩散过程来预测残差噪声并正则化

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the shared representation. By discarding the diffusion branch at test time, ZID-Net integrates the representation-shaping benefits of diffusion supervision into a pure feed-forward architecture. Extensive experiments on synthetic, real-world, and remote-sensing benchmarks demonstrate robust restoration across diverse haze conditions, with particularly strong performance on StateHaze1k, where ZID-Net achieves an average PSNR gain of 3.06 dB while maintaining efficient deterministic inference at 19.35 ms.

Keywords: Image Dehazing, Frequency-spatial decoupling, Zero-inference prior, Diffusion prior, Efficient vision model

1. Introduction

Images captured in hazy weather exhibit severe contrast attenuation, color distortion, and detail loss, as suspended particles weaken direct transmission and introduce scattered airlight. These degradations heavily restrict downstream perception tasks in applications such as autonomous driving [1,2], remote sensing [3,4], and medical diagnostics [5,6]. Although the atmospheric scattering model provides a principled physical formulation, recovering a clear image from a single observation remains highly ill-posed. Furthermore, explicitly estimating the transmission map or atmospheric light is frequently unreliable under dense, non-homogeneous, or real-world haze.

Although recent CNN and transformer-based models have advanced single image dehazing, they still struggle with dense and non-homogeneous haze. Specifically, these architectures inherently entangle the underlying scene semantics with haze patterns, making restoration in heavy-haze scenarios highly unstable. Furthermore, when conventional high-frequency priors (e.g., Laplacian filters) are utilized to preserve structural edges, they lack the

共享表示。通过在测试时丢弃扩散分支，ZID-Net 将扩散监督的表示塑造优势整合到纯前馈架构中。在合成、真实世界和遥感基准上的大量实验表明，该方法能够在多样化的雾霾条件下实现稳健复原，尤其在 StateHaze1k 上表现突出，ZID-Net 取得了 3.06 dB 的平均峰值信噪比增益，同时保持 19.35 ms 的高效确定性推理。

关键词：图像去雾、频空解耦、零推理先验、扩散先验、高效视觉模型

1. 引言

在雾霾天气中拍摄的图像会表现出严重的对比度衰减、色彩失真和细节损失，因为悬浮颗粒会削弱直接透射并引入散射的大气光。这些退化严重限制了自动驾驶 [1,2]、遥感 [3,4]，和医疗诊断 [5,6]等应用中的下游感知任务。尽管大气散射模型提供了一种具有原理性的物理公式，但从单次观测中恢复清晰图像仍然高度不适定。此外，在浓雾、非均匀雾霾或真实世界雾霾条件下，显式估计透射图或大气光通常不可靠。

尽管近年来基于CNN和Transformer的模型推动了单图像去雾的发展，但它们在浓雾和非均匀雾霾场景下仍然表现不佳。具体而言，这些架构本质上会将底层场景语义与雾霾模式纠缠在一起，导致在重雾场景下的复原过程极不稳定。此外，当使用传统的高频先验（如拉普拉斯滤波器）来保留结构边缘时，它们缺乏

mechanism to explicitly suppress the noise amplified by the haze. As a result, corrupted details are inadvertently injected back into the restoration process. While transformer and hybrid designs attempt to address this through global context modeling, they significantly increase computational overhead without fundamentally resolving the underlying feature purity problem.

Recently, diffusion models have demonstrated remarkable capability as generative priors for image restoration. Compared to explicit physical formulations such as the atmospheric scattering model [7], diffusion models excel at learning complex, spatially varying haze distributions in real-world scenarios. However, the iterative reverse sampling process incurs massive inference latency and memory overhead, while its stochasticity leads to inference inconsistency and structural artifacts. This creates a fundamental dilemma: feed-forward networks are computationally efficient but struggle to model severe haze, whereas diffusion-based methods are highly expressive yet prohibitively expensive and unstable at test time.

To address this dilemma, we propose ZID-Net, a frequency-spatial decoupled dehazing network guided by a zero-inference diffusion prior. In the feed-forward backbone, Lightweight Global Context Blocks (LGCBs) are embedded at the bottleneck to establish long-range spatial dependencies. Simultaneously, a Channel-Spatial Laplacian Mask (CSLM) dynamically filters haze-amplified noise from color Laplacian residuals, extracting purified chromatic and structural details. A Dynamic Feature Arbitration Block (DFAB) then adaptively fuses the semantic, structural, and upsampled decoding streams for reconstruction. To incorporate generative priors without increasing inference cost, we attach a Zero-Inference Prior Propagation Head (ZI-PPH). During

一种能够显式抑制由雾霾放大的噪声的机制。结果，被破坏的细节会不经意地被重新注入到复原过程中。虽然Transformer和混合设计试图通过全局上下文建模来解决这一问题，但它们显著增加了计算开销，却并未从根本上解决底层特征纯度的问题。

近年来，扩散模型作为图像复原的生成先验展现出了卓越的能力。与大气散射模型等显式物理公式相比，[7]，扩散模型更擅长学习真实世界中复杂、空间变化的雾霾分布。然而，迭代反向采样过程会带来巨大的推理延迟和内存开销，同时其随机性会导致推理不一致和结构性伪影。这就形成了一个根本性困境：前馈网络计算高效，但难以应对严重雾霾；而基于扩散的方法虽然表达能力强，但在测试时却极其昂贵且不稳定。

为了解决这一难题，我们提出了ZID-Net，一种由零推理扩散先验引导的频率-空间解耦去雾网络。在前馈主干网络中，轻量级全局上下文模块（LGCBs）被嵌入瓶颈层以建立长程空间依赖。同时，通道-空间拉普拉斯掩码（CSLM）动态过滤颜色拉普拉斯残差中的雾放大噪声，提取纯净的色彩信息和结构细节。然后，动态特征仲裁模块（DFAB）自适应地融合语义、结构和上采样解码流以进行重建。为了在不增加推理成本的情况下引入生成先验，我们附加了一个零推理先验传播头（ZI-PPH）。在

training, this conditional diffusion branch predicts haze-residual noise, explicitly guiding the shared bottleneck to learn degradation-aware features. At test time, the ZI-PPH is entirely discarded, preserving a strictly one-pass inference architecture.

- The main contributions of ZID-Net can be summarized as follows:
- (1) We propose a framework that decouples diffusion supervision from inference for single image dehazing. By restricting diffusion priors to the training phase, our method improves restoration performance without compromising feed-forward inference speed.
 - (2) We design a frequency-spatial decoupled architecture for direct inference. This structure integrates LGCBs for global semantic aggregation, CSLM for color Laplacian residual purification, and dynamic feature arbitration to achieve high-quality and efficient image restoration.
 - (3) We introduce a zero-inference conditional diffusion prior exclusively during the training phase to inject generative supervision into deterministic inference. By predicting haze-residual noise during training, it explicitly guides the shared backbone to model the haze distribution.

2. Related Work

Image dehazing has evolved significantly, transitioning from early physical priors to end-to-end deep networks, and most recently, to diffusion models. In this section, we briefly review these three types of methods.

2.1. Physical Prior-based Image Dehazing

Early approaches relied heavily on the atmospheric scattering model, seeking to recover the transmission map and global atmospheric light through statistical priors. Representative methods include the Dark Channel Prior (DCP) [8] and the Color Attenuation Prior [9]. However, constrained by their

训练期间，该条件扩散分支预测雾残差噪声，明确引导共享瓶颈学习退化感知特征。在测试时，ZI-PPH被完全丢弃，保持了严格的单遍推理架构。

ZID-Net的主要贡献可总结如下：（1）我们提出了一种将扩散监督与推理解耦的框架，用于单图像去雾。通过将扩散先验限制在训练阶段，我们的方法在提升恢复性能的同时，不影响前馈推理速度。

（2）我们设计了一种面向直接推理的频域-空间解耦架构。该结构集成了用于全局语义聚合的LGCBs、用于色彩拉普拉斯残差净化的CSLM以及动态特征仲裁，以实现高质量且高效的图像复原。

（3）我们仅在训练阶段引入零推理条件扩散先验，将生成式监督注入确定性推理。通过在训练过程中预测雾残差噪声，它显式地引导共享骨干网络对雾气分布进行建模。

2. 相关工作

图像去雾已取得显著发展，从早期基于物理先验的方法，过渡到端到端深度网络，再到最近的扩散模型。本节中，我们简要回顾这三类方法。

2.1. 基于物理先验的图像去雾

早期方法严重依赖大气散射模型，试图通过统计先验来恢复透射图和全局大气光。代表性方法包括暗通道先验 (DCP)[8] 和颜色衰减先验 [9]。然而，受限于其

handcrafted nature and simplified physical assumptions, these formulations struggle to adapt to complex real-world hazy conditions. Consequently, they frequently produce severe color distortion and halo artifacts when processing regions with dense haze, bright skies, or substantial depth variations.

2.2. End-to-End Image Dehazing

To overcome the fragility of empirical priors, early CNN architectures like DehazeNet [10] and AOD-Net [11] introduced trainable parameter mappings. Subsequent works bypassed physical formulations entirely, advancing to fully end-to-end networks like FFA-Net [12]. Recent models further extend this pipeline with specific auxiliary mechanisms: C2PNet [13] employs contrastive learning to explicitly separate clear and hazy feature representations, DCMNet [14] reintroduces depth maps to guide physical scene understanding, DEA-Net [15] focuses on restoring high-frequency local details, and SGDN [16] utilizes the YCbCr [17] color space to preserve structural integrity independent of color distortion. Recent methods also explore feature decoupling and knowledge transfer: DCE-Net [18] leverages dual-frequency knowledge for detail and content restoration, ZRID-Net [19] employs self-decoupling and reverse knowledge transfer for zero-reference dehazing, and TSID-Net [20] combines style transfer with contrastive knowledge transfer. While improving visual quality, many learning-based dehazing methods still rely heavily on localized convolutional operations, limiting their ability to model non-homogeneous haze and causing feature entanglement where skip-connections bypass hazy artifacts into the decoder.

To overcome the limited receptive fields of CNNs, Transformers have been widely adopted for global context modeling. For instance, DehazeFormer [21]

手工设计特性和简化的物理假设，这些公式难以适应复杂的真实世界雾霾条件。因此，它们在处理浓雾、明亮天空或深度变化较大的区域时，经常产生严重的色彩失真和光晕伪影。

2.2. 端到端图像去雾

为克服经验先验的脆弱性，DehazeNet [10] 和 AOD-Net [11] 等早期 CNN 架构引入了可训练参数映射。后续工作完全绕开物理公式，发展为 FFA-Net [12]等全端到端网络。近期模型进一步用特定辅助机制扩展这一流程：C2PNet [13] 采用对比学习来显式分离清晰与有雾特征表示，DCMNet[14] 重新引入深度图来引导物理场景理解，DEA-Net[15]专注于恢复高频局部细节，SGDN [16] 利用 YCbCr [17] 色彩空间在不受色彩失真影响的情况下保持结构完整性。近期方法还探索了特征解耦与知识迁移：DCE-Net [18]利用双频知识进行细节与内容恢复，ZRID-Net [19] 采用自解耦和逆向知识迁移实现零参考去雾，TSID-Net [20] 将风格迁移与对比知识迁移相结合。尽管视觉质量有所提升，许多基于学习的去雾方法仍严重依赖局部卷积运算，这限制了它们对非均匀雾霾的建模能力，并导致跳跃连接绕过雾霾伪影进入解码器时产生特征纠缠。

为了克服卷积神经网络有限的感受野，Transformer已被广泛用于全局上下文建模。例如，DehazeFormer [21]

modifies standard normalization within self-attention to better capture non-local haze distributions. However, since traditional self-attention scales quadratically with image resolution, subsequent models like the MB-TaylorFormer [22,23] series approximate the softmax operation via Taylor expansion to achieve linear complexity. Furthermore, UDPNet [24] employs cross-attention to integrate physical depth cues with visual representations, using depth priors to restore structures under spatially varying haze. Despite these optimizations, Transformer architectures inherently demand higher inference overhead than standard CNNs. To balance computational efficiency with global context, we propose LGCBs. Instead of relying on computationally heavy self-attention variations, LGCBs efficiently establish long-range spatial dependencies, achieving robust global reasoning with the deterministic efficiency of CNNs.

2.3. Diffusion Models in Image Dehazing

Diffusion models have recently emerged as a powerful tool in low-level vision by providing strong generative priors. For instance, DiffIR [25] demonstrates the efficacy of diffusion processes in general image restoration. Building on this, recent studies adapt diffusion models specifically for the dehazing task. FGPS [26], for example, integrates frequency-domain guidance into reverse sampling to enhance structural recovery. Recent studies further combine diffusion restoration with optimal-transport-based unpaired training [27] or degradation-aware prompts derived from pretrained vision-language models [28]. Concurrently, generative models are also being exploited to address the discrepancy between synthetic and real-world haze, as seen in methods like *Learning Hazing to Dehazing* [29], which synthesize realistic hazy images for better supervision.

修改了自注意力中的标准归一化，以更好地捕获非局部的雾霾分布。然而，由于传统的自注意力随图像分辨率呈二次方增长，后续模型如 MB-TaylorFormer [22,23]系列通过泰勒展开近似softmax操作以达成线性复杂度。此外，UDPNet [24]采用交叉注意力将物理深度线索与视觉表示相融合，利用深度先验在空间变化雾条件下恢复结构。尽管有了这些优化，Transformer架构本质上仍比标准CNN需要更高的推理开销。为在计算效率与全局上下文之间取得平衡，我们提出了LGCBs。LGCBs不依赖计算繁重的自注意力变体，而是高效地建立长程空间依赖，以卷积神经网络的确定性效率实现稳健的全局推理。

2.3. Diffusion Models in Image Dehazing

扩散模型近年来凭借其强大的生成先验，已成为低级视觉领域的一种有力工具。例如，DiffIR [25]展示了扩散过程在通用图像复原中的有效性。在此基础上，近期研究将扩散模型专门应用于去雾任务。例如，FGPS [26]，将频域引导融入逆向采样，以增强结构恢复。进一步的研究还将扩散复原与基于最优传输的非配对训练[27] 或源自预训练视觉语言模型的退化感知提示 [28]相结合。与此同时，生成模型也被用于解决合成雾霾与现实世界雾霾之间的差异，如*Learning Hazing to Dehazing* [29], 等方法通过合成真实雾图来提供更好的监督。

While these approaches advance the field, they face distinct limitations. Employing diffusion models directly for restoration incurs high iterative sampling costs and training-inference inconsistency. Unlike these methods, ZID-Net explores the conditional diffusion branch exclusively as a training regularizer. By predicting residual noise during training, it explicitly guides the shared backbone to model the underlying haze distribution. Because this branch is completely discarded during testing, our method benefits from the diffusion prior, achieving deterministic inference without the latency of iterative sampling.

3. Proposed Method

In this section, we detail the architecture of the proposed ZID-Net. As shown in Fig. 1, our framework adopts a decoupled design: a deterministic network for direct inference and an auxiliary diffusion prior branch solely for training. First, Section 3.1 introduces the frequency-spatial decoupled dehazing backbone, which extracts global semantic contexts while preserving high-frequency chromatic details. Next, Section 3.2 presents the Zero-Inference Prior Propagation Head (ZI-PPH). This training-only auxiliary branch utilizes a conditional diffusion process to guide the backbone in modeling the haze distribution, enabling deterministic inference without reverse-sampling latency. Finally, the objective functions are detailed in Section 3.3.

尽管这些方法推动了该领域的发展，但它们也面临着各自的局限。直接使用扩散模型进行图像复原会带来高昂的迭代采样成本和训练-推理不一致性。与这些方法不同，ZID-Net 仅将条件扩散分支用作训练正则化器。通过在训练过程中预测残差噪声，它明确引导共享骨干网络对底层雾气分布进行建模。由于该分支在测试阶段被完全舍弃，我们的方法在受益于扩散先验的同时，实现了确定性推理，避免了迭代采样带来的延迟。

3. 提出的方法

在本节中，我们将详细介绍所提出的 ZID-Net 的架构。如图 1 所示，我们的框架采用了解耦设计：一个用于直接推理的确定性网络和一个仅用于训练的辅助扩散先验分支。首先，第 3.1 节介绍了频率-空间解耦去雾主干网络，该网络在保留高频色彩细节的同时提取全局语义上下文。接着，第 3.2 节介绍了零推理先验传播头（ZI-PPH）。该仅训练辅助分支利用条件扩散过程来引导骨干网络对雾气分布进行建模，从而在无需反向采样延迟的情况下实现确定性推理。最后，目标函数在第 3.3 节中详细说明。

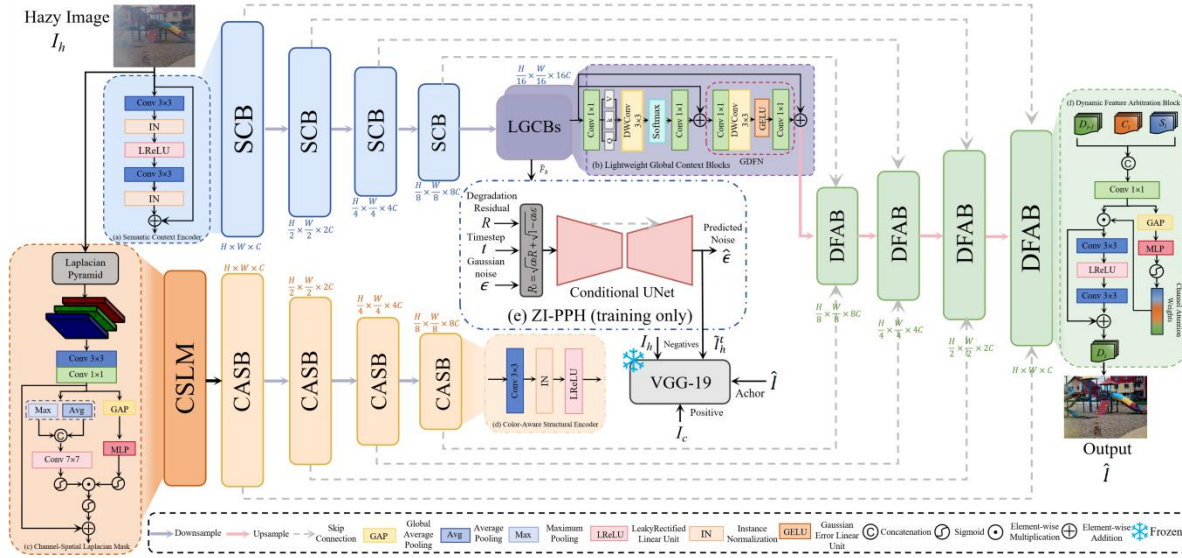


Fig. 1. Overall architecture of the proposed ZID-Net. The network adopts a dual-stream deterministic backbone regularized by a diffusion-based prior. (a) The Semantic Context Block (SCB) extracts multi-scale context features from the input hazy image. (b) The LGCBS aggregate scene-level deep semantic features at the bottleneck stage. (c) The CSLM extracts and adaptively filters high-frequency physical residuals using a Laplacian pyramid. (d) The Color-Aware Structural Block (CASB) extracts high-frequency structural features. (e) The ZI-PPH regularizes the bottleneck features via a conditional U-Net during training and is detached during testing. (f) The DFAB fuses multiscale semantic and structural features to decode and reconstruct the final clear image.

3.1. Frequency-Spatial Decoupled Dehazing Backbone

As depicted in the main pathway of Fig. 1, the feed-forward backbone employs a dual-stream architecture to process the input. Given a hazy image I_h , we first compute a color high-frequency residual I_{hf}^h to decouple the frequency and spatial representations. Then, I_h is fed into a semantic context encoder for global semantic modeling, while I_{hf}^h is routed to a color-aware structural encoder for screened high-frequency extraction. Finally, the DFAB decoder adaptively fuses these dual-branch features with the upsampled states to reconstruct the clear image.

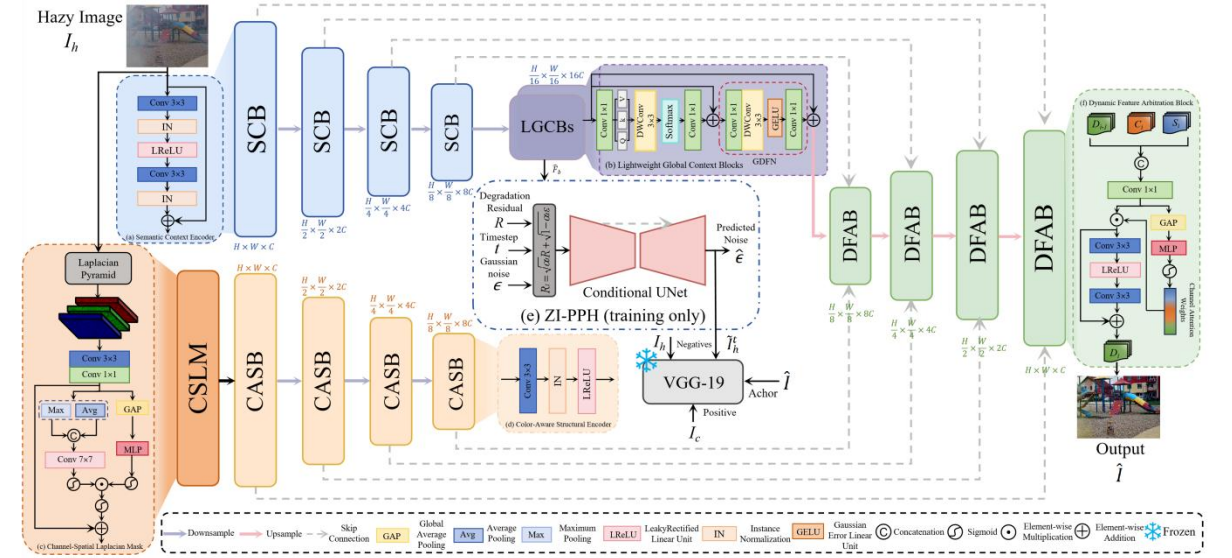


图 1. 本文提出的 ZID-Net 总体架构。该网络采用基于扩散的先验正则化的双流确定性主干网络。(a) 语义上下文块 (SCB) 从输入的有雾图像中提取多尺度上下文特征。(b) LGCBS 在瓶颈阶段聚合场景级深层语义特征。(c) CSLM 使用拉普拉斯金字塔提取并自适应过滤高频物理残差。(d) 颜色感知结构块 (CASB) 提取高频结构特征。(e) ZI-PPH 在训练期间通过条件 U-Net 正则化瓶颈特征，并在测试期间被分离。(f) DFAB 融合多尺度语义与结构特征，以解码并重建最终的清晰图像。

3.1. 频域-空间解耦去雾骨干网络

如图 1 的主通路所示，前馈主干网络采用双流架构来处理输入。给定有雾图像 I_h ，我们首先计算颜色高频残差 I_{hf}^h 以解耦频率与空间表示。然后， I_h 被送入语义上下文编码器进行全局语义建模，而 I_{hf}^h 则被路由至颜色感知结构编码器进行筛选高频提取。最后，DFAB 解码器将这些双分支特征与上采样状态自适应融合，以重建清晰图像。

The motivation for this frequency-spatial separation arises from the different spatial behaviors of haze-related variations and scene structures, as illustrated in Fig. 2. Illumination changes and atmospheric effects usually extend over broad regions, resulting in low-frequency variations in image intensity and color. In contrast, scene boundaries, textures, and chromatic transitions occur at local spatial scales and are mainly represented by high-frequency components. Spatially varying haze may also introduce high-frequency fluctuations near regions with abrupt degradation changes, making it necessary to preserve useful structural responses while suppressing unreliable ones. Fig. 2 shows that the high-frequency components of the independent R, G, and B channels maintain strong linear correlations with their clear-image counterparts, whereas a single-channel representation loses part of this chromatic information. These observations motivate using the color high-frequency residual as an independent structural input, while the semantic branch focuses on global haze-related variations.

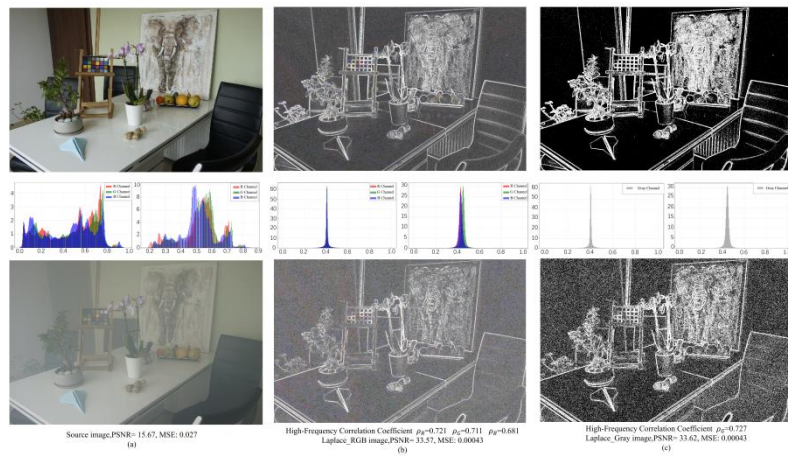


Fig. 2. Comparative analysis of high-frequency characteristics between clean and hazy images. (a) Source images with intensity histograms (left: clean; right: hazy) and metrics. (b) RGB-channel Laplacian maps, high-frequency correlations, histograms (left: clean; right: hazy), and metrics. (c) Single-channel Laplacian maps, high-frequency correlations, histograms (left: clean; right: hazy),

这种频率-空间分离的动机源于雾相关变化与场景结构在空间行为上的差异，如图2所示。光照变化和大气效应通常覆盖较广区域，导致图像强度和颜色出现低频变化。相比之下，场景边界、纹理和色彩过渡发生在局部空间尺度上，主要由高频分量表示。空间变化雾也可能在退化突变区域附近引入高频波动，因此有必要在抑制不可靠结构响应的同时保留有用的结构响应。图2显示，独立R、G、B通道的高频分量与其清晰图像对应部分保持强线性相关性，而单通道表示会丢失部分色彩信息。这些观察结果促使我们将颜色高频残差作为独立的结构输入，而语义分支则专注于全局雾相关变化。

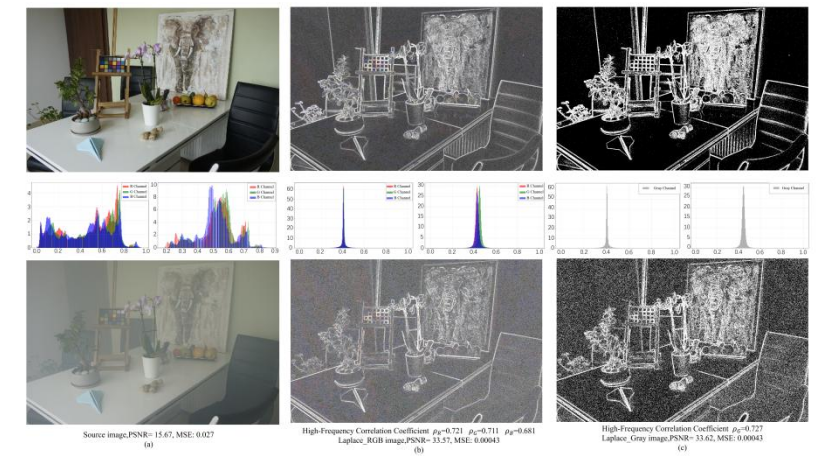


图 2. 清晰图像与有雾图像高频特性的对比分析。(a) 带有强度直方图的源图像（左：清晰；右：有雾）及指标。(b) RGB通道拉普拉斯图、高频相关性、直方图（左：清晰；右：有雾）及指标。(c) 单通道拉普拉斯图、高频相关性、直方图（左：清晰；右：有雾），

and metrics.

Semantic Context Encoder with LGCB. Standard convolutions expand the receptive field only gradually, which is restrictive when haze thickness, airlight, and color bias correlate across distant regions. The semantic context encoder therefore applies a 3×3 projection with Instance Normalization and LeakyReLU, followed by four stages of a residual block and a stride-2 convolution to produce the multiscale hierarchy $\{S_0, S_1, S_2, S_3\}$. At the deepest stage, the spatial dimensions are compressed to 1/16 of the original resolution, resulting in a bottleneck feature $F_b \in \mathbb{R}^{16C \times \frac{H}{16} \times \frac{W}{16}}$. Defining $N = \frac{HW}{256}$ as the spatial cardinality at this resolution, we use Lightweight Global Context Blocks (LGCBs) on F_b to aggregate scene-level context. This formulation avoids the quadratic computational burden inherent in conventional spatial self-attention.

For an input feature $X \in \mathbb{R}^{16C \times \frac{H}{16} \times \frac{W}{16}}$, LGCB first generates the query, key, and value tensors through a fused projection block. Specifically, a point-wise convolution $W_p(\cdot)$ expands the channel dimension to $48C$, followed by a depth-wise convolution $W_d(\cdot)$ to encode spatial context. The resulting tensor is then split equally along the channel dimension:

$$[Q, K, V] = \text{Split}(W_d(W_p(X))) \quad (1)$$

where $Q, K, V \in \mathbb{R}^{16C \times \frac{H}{16} \times \frac{W}{16}}$. Rather than computing spatial attention over $N \times N$ locations, LGCB reshapes these tensors into $\mathbb{R}^{16C \times N}$ and performs transposed attention in the channel space. Let $\bar{Q}$ and $\bar{K}$ denote the l_2 -normalized representations of Q and K along the spatial dimension, and $\bar{V}$ be the reshaped value tensor. The attended representation is computed as:

$$X' = X + W_o \left(\text{Softmax}(\bar{K} \bar{Q}^\top / \tau) \bar{V} \right) \quad (2)$$

和指标。

具有LGCB的语义上下文编码器。标准卷积只能逐步扩大感受野，当雾霾厚度、大气光和颜色偏差在远距离区域之间具有相关性时，这种限制尤为明显。因此，语义上下文编码器先应用一个 3×3 带有实例归一化和 LeakyReLU的投影，随后通过四个阶段的残差模块和步长2卷积来生成多尺度层级 $\{S_0, S_1, S_2, S_3\}$ 。在最深阶段，空间维度被压缩到 1/16原始分辨率的/16，产生瓶颈特征 $F_b \in \mathbb{R}^{16C \times \frac{H}{16} \times \frac{W}{16}}$ 。定义 $N = \frac{HW}{256}$ 为该分辨率下的空间基数，我们对 F_b 使用轻量级全局上下文模块（LGCBs）来聚合场景级上下文。这种设计避免了传统空间自注意力固有的二次计算负担。



Figure 1: Architecture of the proposed LGCB. For an input feature $X \in \mathbb{R}^{16C \times \frac{H}{16} \times \frac{W}{16}}$, LGCB first generates the query, key, and value tensors through a fused projection block. Specifically, a point-wise convolution $W_p(\cdot)$ expands the channel dimension to $48C$, followed by a depth-wise convolution $W_d(\cdot)$ to encode spatial context. The resulting tensor is then split equally along the channel dimension:

$$[Q, K, V] = \text{Split}(W_d(W_p(X))) \quad (1)$$

其中 $Q, K, V \in \mathbb{R}^{16C \times \frac{H}{16} \times \frac{W}{16}}$ 。LGCB 并非在 $N \times N$ 个位置上计算空间注意力，而是将这些张量重塑为 $\mathbb{R}^{16C \times N}$ ，并在通道空间中执行转置注意力。设 Q 和 K 分别表示 Q 和 K 沿空间维度的 l_2 归一化表示， V 为重塑后的值张量。则注意力表示的计算方式为：

$$X' = X + W_o \left(\text{Softmax}(\bar{K} \bar{Q}^\top / \tau) \bar{V} \right) \quad (2)$$

where τ is a learnable temperature, $\text{Softmax}(\cdot)$ is applied along the channel dimension to compute channel-wise attention weights, and $W_o(\cdot)$ is a 1×1 output projection.

By computing attention across the $16C$ channel dimensions, the network enables each channel to interact through scene-wide responses rather than being restricted to local windows.

To further refine this mixed representation, LGCB employs a Gated Depth-wise Feed-Forward Network (GDFN). The GDFN expands the channel dimension, applies a depth-wise convolution, and splits the intermediate feature into two parallel paths to perform a gating operation:

$$\widehat{X} = X' + W_2(GELU(U_1) \odot U_2) \quad (3)$$

where $[U_1, U_2] = \text{Split}(W_d(W_1(X')))$, and W_1, W_2 are point-wise linear projections.

By computing attention across the channel dimension, the computational complexity of LGCB scales linearly with respect to the spatial resolution N . Let C_b denote the channel capacity at the bottleneck. The dominant cost of LGCB is $\mathcal{O}(C_b^2 N)$, which provides a significant efficiency advantage over the $\mathcal{O}(N^2 C_b)$ quadratic complexity of conventional spatial self-attention. This linear scaling is particularly advantageous for high-resolution inputs. For example, when increasing the input resolution from 512×512 to 1024×1024 , the computational overhead of traditional spatial attention scales quadratically, often leading to memory exhaustion. In contrast, LGCB maintains a manageable linear increase. Furthermore, by applying LGCBs exclusively at the maximally

其中 τ 是可学习的温度参数, $\text{Softmax}(\cdot)$ 沿通道维度应用, 以计算通道注意力权重, $W_o(\cdot)$ 是 1×1 输出投影。

通过在 $16C$ 个通道维度上计算注意力, 网络使每个通道能够通过场景级响应进行交互, 而非局限于局部窗口。

为了进一步细化这种混合表示, LGCB采用了门控逐深度前馈网络 (GDFN)。该GDFN扩展通道维度, 应用深度卷积, 并将中间特征分成两条并行路径以执行门控运算:

$$\widehat{X} = X' + W_2(GELU(U_1) \odot U_2) \quad (3)$$

其中 $[U_1, U_2] = \text{Split}(W_d(W_1(X')))$, 且 W_1, W_2 为逐点线性投影。

通过在通道维度上计算注意力, LGCB 的计算复杂度相对于空间分辨率 N 呈线性缩放。设 C_b 表示瓶颈层的通道容量。LGCB 的主要成本为 $\mathcal{O}(C_b^2 N)$, 这相对于传统空间自注意力的 $\mathcal{O}(N^2 C_b)$ 二次复杂度具有显著的效率优势。这种线性缩放对于高分辨率输入尤为有利。例如, 当输入分辨率从 512×512 提高到 1024×1024 时, 传统空间注意力的计算开销呈二次增长, 往往导致内存耗尽。相比之下, LGCB 保持可控的线性增长。此外, 通过仅在最大

compressed bottleneck layer, ZID-Net achieves high-density scene-wide semantic aggregation with minimal computational overhead. The final semantic bottleneck output is denoted by $\widehat{F}_b$.

Color-Aware Structural Encoder with CSLM. Motivated by this analysis and the observations in Fig. 2, the Channel-Spatial Laplacian Mask (CSLM) is designed to extract and purify color high-frequency residuals. The module initially extracts a single-scale color residual I_{hf}^h from the hazy input I_h :

$$I_{hf}^h = I_h - Up(Down(G(I_h))) \quad (4)$$

where $G(\cdot)$ is a 5×5 Gaussian filter. Unlike conventional grayscale edge extraction, this operation preserves the channel-specific chromatic structures identified in our analysis.

To map this physical residual into the feature space, CSLM employs a shallow projection followed by a 1×1 color adaptation layer. To suppress haze-amplified fluctuations while preserving reliable edges, the module functions as a dynamic gating mechanism. It computes a channel descriptor M_c via global average pooling and an MLP, and a spatial descriptor M_s via a 7×7 convolution over channel-wise pooling responses. The intermediate feature X is then adaptively filtered and reinforced via a residual connection:

$$\widetilde{X} = X + X \odot \sigma(M_c \odot M_s) \quad (5)$$

where $\sigma(\cdot)$ is the sigmoid function. The filtered feature $\widetilde{X}$ then passes through four stride-2 convolutional stages to form the structural hierarchy $\{C_0, C_1, C_2, C_3\}$.

压缩瓶颈层上应用 LGCBs, ZID-Net 以最小的计算开销实现了高密度场景级语义聚合。最终语义^瓶颈层输出记为 F_b 。

采用CSLM的颜色感知结构编码器。受此分析及图2中观察结果的启发,设计了通道-空间拉普拉斯掩码(CSLM)来提取和净化色彩高频残差。该模块首先从雾霾输入 I_h 中提取单尺度色彩残差 I_{hf}^h :

$$I_{hf}^h = I_h - Up(Down(G(I_h))) \quad (4)$$

其中 $G(\cdot)$ 是一个 5×5 高斯滤波器。与传统的灰度边缘提取不同, 此操作保留了我们分析中识别出的通道特定色彩结构。

为了将该物理残差映射到特征空间, CSLM采用浅层投影, 随后接入一个 1×1 色彩适配层。为了抑制雾霾放大的波动, 同时保留可靠边缘, 该模块起到动态门控机制的作用。它通过全局平均池化和多层感知机计算通道描述符 M_c , 并在逐通道池化响应上通过 7×7 卷积计算空间描述符 M_s 。随后中间特征 X 通过残差连接被自适应滤波并增强:

$$\widetilde{X} = X + X \odot \sigma(M_c \odot M_s) \quad (5)$$

其中 $\sigma(\cdot)$ 是 sigmoid 函数。滤波后的特征 X 随后经过四个步长为2的卷积阶段, 形成结构化层次 $\{C_0, C_1, C_2, C_3\}$ 。

Dynamic Feature Arbitration Block (DFAB) Decoder. The decoder reconstructs the clear image by progressively fusing dual-encoder representations with the upsampled decoder states. Starting from the bottleneck $\hat{F}_b$, each stage employs a DFAB to dynamically arbitrate between the semantic skip feature S_i , the structural feature C_i , and the upsampled state $\tilde{D}_i = Up(D_{i+1})$.

Rather than relying on standard addition or concatenation, DFAB performs content-adaptive fusion via channel-wise reweighting. The three-stream input is first concatenated and compressed into a shared representation $Z_i = W_p([\tilde{D}_i, S_i, C_i])$ via a 1×1 convolution. DFAB then computes Squeeze-and-Excitation (SE) weights to dynamically recalibrate the channel responses based on the fused content. The recalibrated feature is further refined by a residual block to produce the current decoder state:

$$D_i = ResBlock(Z_i \odot \sigma(W_2 \delta(W_1 GAP(Z_i)))), \quad i \in \{0,1,2,3\} \quad (6)$$

where $GAP(\cdot)$ is global average pooling, and $\delta(\cdot)$ denotes the ReLU activation.

By utilizing SE-style channel reweighting and residual refinement independently at each scale, DFAB balances structural and semantic information effectively, avoiding the computational overhead of explicit spatial or cross-attention mechanisms before the final 3×3 prediction layer.

The semantic and structural branches play complementary roles in image reconstruction. The semantic branch models global illumination and long-range context at the bottleneck, whereas the Laplacian branch provides local edges,

动态特征仲裁模块（DFAB）解码器。解码器通过逐步将双编码器表示与上采样后的解码器状态融合来重建清晰图像。从瓶颈层 $\hat{F}_b$ 开始，每一级使用一个 DFAB 在语义~跳跃特征 S_i 、结构特征 C_i 和上采样状态 $D_i = Up(D_{i+1})$ 之间进行动态仲裁。

DFAB并非依赖标准的相加或拼接，而是通过逐通道重加权执行内容自适应融合。三流输入首先被拼接并经一个 1×1 卷积压缩为共享表示 $Z_i = W_p([D_i, S_i, C_i])$ 。随后，DFAB计算压缩-激励（SE）权重，以根据融合内容动态重新校准通道响应。重校准后的特征进一步由残差模块细化，生成当前解码器状态：

$$D_i = ResBlock(Z_i \odot \sigma(W_2 \delta(W_1 GAP(Z_i)))), \quad i \in \{0,1,2,3\} \quad \text{其6}$$

中 $GAP(\cdot)$ 为全局平均池化， $\delta(\cdot)$ 表示ReLU激活函数。

通过在每个尺度上独立利用SE式通道重加权和残差细化，DFAB有效平衡了结构与语义信息，避免了在最终 3×3 预测层之前引入显式空间注意力或交叉注意力机制所带来的计算开销。

语义分支和结构分支在图像重建中扮演互补的角色。语义分支在瓶颈层建模全局光照和长程上下文，而拉普拉斯分支提供局部边缘、

textures, and color details. CSLM filters unreliable high-frequency responses caused by haze, and DFAB adaptively combines the semantic, structural, and decoder features at each scale to reconstruct regions with different haze conditions.

3.2. Zero-Inference Prior Propagation Head

Methods grounded in the atmospheric scattering model [10,11] often suffer from error accumulation when explicitly estimating the transmission map and atmospheric light under restrictive physical assumptions. Conversely, while purely data-driven feed-forward dehazers bypass these physical bottlenecks and offer high computational efficiency, relying entirely on standard reconstruction losses leaves their degradation modeling weakly constrained, particularly under dense or spatially varying haze. While diffusion models offer strong generative priors, their direct application incurs heavy iterative sampling costs. To bypass this overhead, we introduce a training-only auxiliary branch termed the Zero-Inference Prior Propagation Head (ZI-PPH). As illustrated in Fig. 3, this module operates entirely within the haze-residual domain.

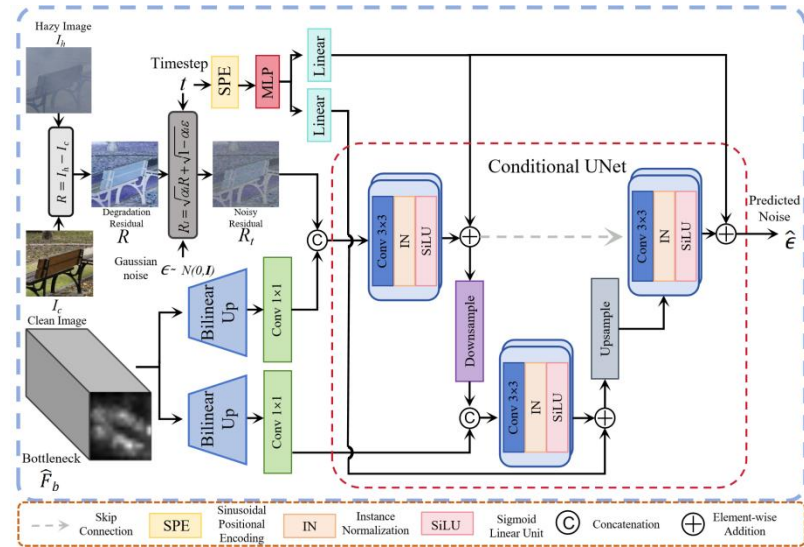


Fig. 3. Detailed architecture of the ZI-PPH branch. This training-only auxiliary module regularizes

纹理和颜色细节。CSLM过滤由雾霾引起的不可靠高频响应，DFAB在每个尺度上自适应地结合语义、结构和解码器特征，以重建具有不同雾霾条件的区域。

3.2. Zero-Inference Prior Propagation Head

基于大气散射模型 [10,11] 的方法在限制性物理假设下显式估计透射图和大气光时，往往会遭受误差累积的困扰。相反，虽然纯数据驱动的前馈去雾器绕开了这些物理瓶颈并提供了较高的计算效率，但完全依赖标准重建损失使其退化建模约束较弱，特别是在浓雾或空间变化雾条件下。虽然扩散模型提供了强大的生成先验，但其直接应用会带来高昂的迭代采样成本。为了规避这一开销，我们引入了一个仅训练辅助分支，称为零推理先验传播头（ZI-PPH）。如图3所示，该模块完全在雾霾残差域中运行。

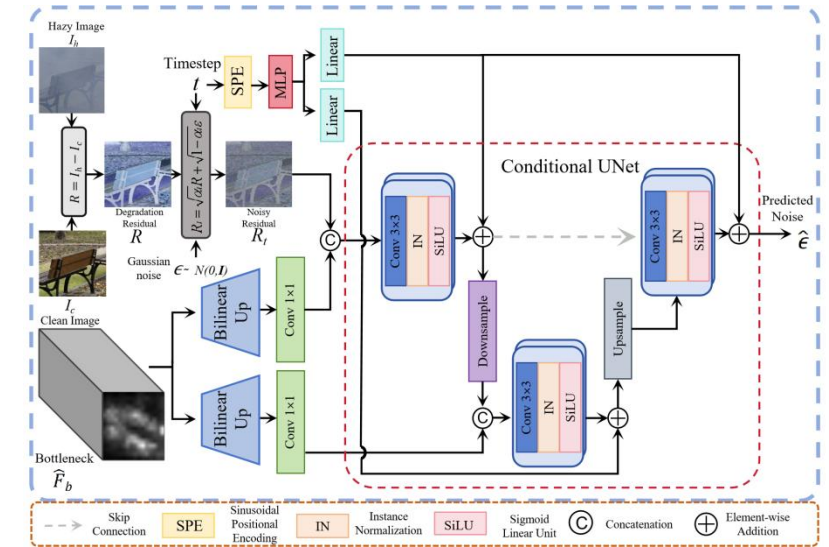


Fig. 3. Detailed architecture of the ZI-PPH branch. This training-only auxiliary module regularizes

the shared bottleneck features $\hat{F}_b$ via a conditional U-Net. The noisy physical residual R_t and $\hat{F}_b$ are concatenated at the input level, with the timestep t (encoded via SPE and MLP) additively injected across the network. This branch predicts the added noise $\hat{\epsilon}$ to provide auxiliary degradation-aware supervision to the shared representation.

Given a hazy-clean pair (I_h, I_c) , we first define the degradation residual as

$R = I_h - I_c$. For a randomly sampled diffusion timestep t , the forward perturbation process constructs a noisy residual:

$$R_t = \sqrt{\alpha_t} R + \sqrt{1 - \alpha_t} \epsilon \quad (7)$$

where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ and α_t is the cumulative product of the variance schedule.

Unlike conventional diffusion models that sample t uniformly from $[0, T]$, we introduce a batch-wise severity-adaptive sampling strategy. For the i -th sample in a mini-batch, we calculate the mean absolute error of its residual, $s_i = \text{mean}(|I_{h,i} - I_{c,i}|)$, to estimate the haze concentration. This is normalized within the batch to yield a relative severity score $s_{norm,i} \in [0,1]$. Using this score, we dynamically constrain the upper bound of the sampled timestep:

$$T_i^{cap} = T_{low} + \gamma \cdot s_{norm,i} \cdot (T - T_{low}) \quad (8)$$

where T_{low} is a predefined baseline and γ controls the adaptive strength. The timestep is then uniformly drawn from $[0, T_i^{cap}]$. Consequently, lighter haze samples are assigned smaller timesteps to preserve high-frequency details, while heavier haze samples are exposed to a broader noise range to capture robust structural priors.

$\hat{\epsilon}$ 共享瓶颈特征 F_b 通过条件U-Net。带噪物理残差 R_t 和 F_b 在输入层拼接，时间步 t （通过正弦位置编码和多层感知机编码）以加性方式注入整个网络。该分支预测添加的噪声 ϵ ，为共享表示提供辅助的退化感知监督。

给定一对雾霾-清晰图像 (I_h, I_c) ，我们首先将退化残差定义为 $R = I_h - I_c$ 。对于一个随机采样的扩散时间步 t ，前向扰动过程会构造一个噪声残差：

$$R_t = \sqrt{\alpha_t} R + \sqrt{1 - \alpha_t} \epsilon \quad (7)$$

其中 $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ ，且 α_t 为方差调度的累积乘积。

与从 $[0, T]$ 中均匀采样 t 的传统扩散模型不同，我们引入了一种逐批次严重程度自适应采样策略。对于小批次中的第*i*个样本，我们计算其残差的平均绝对误差， $s_i = \text{mean}(|I_{h,i} - I_{c,i}|)$ ，以估计雾浓度。该误差在批次内进行归一化，得到相对严重程度分数 $s_{norm,i} \in [0,1]$ 。利用该分数，我们动态约束采样时间步的上限：

$$T_i^{cap} = T_{low} + \gamma \cdot s_{norm,i} \cdot (T - T_{low}) \quad (8)$$

其中 T_{low} 是一个预定义的基线， γ 控制自适应强度。然后时间步从 $[0, T_i^{cap}]$ 中均匀采样。因此，较轻的雾霾样本被分配较小的时间步以保留高频细节，而较重的雾霾样本则暴露于更宽的噪声范围以捕获鲁棒的结构先验。

Rather than executing a full reverse chain, ZI-PPH predicts the injected perturbation ϵ conditioned on the shared semantic bottleneck representation $\hat{F}_b$. This conditioning turns the diffusion objective into an auxiliary feature-level regularizer: accurate perturbation estimation across varying timesteps requires $\hat{F}_b$ to preserve structure-consistent semantic information while remaining informative about haze-dependent degradation. To transfer this degradation-aware guidance effectively through the conditional U-Net, $\hat{F}_b$ is spatially aligned and fused at two distinct semantic scales.

At the input level, $\hat{F}_b$ is bilinearly upsampled to the original resolution, compressed via a 1×1 projection, and concatenated with the noisy residual R_t . This early injection provides pixel-aligned global context, enabling the shallow convolutional layers to distinguish between the spatial haze distribution and the underlying structural details.

Deeper in the network, $\hat{F}_b$ is upsampled to the half-resolution scale and additively integrated into the intermediate encoder representations. This secondary fusion reinforces the semantic prior at a higher level of abstraction, preventing the conditioning signal from decaying as features undergo progressive spatial downsampling.

Alongside this multi-scale spatial conditioning, the diffusion timestep t is encoded via sinusoidal embeddings and a multi-layer perceptron, then added across the U-Net hierarchy. This temporal modulation ensures that all layers maintain awareness of the current perturbation level along the denoising trajectory. Formally, the auxiliary branch predicts:

ZI-PPH 并非执行完整的逆向链，而是以共享语义瓶颈表示 $\epsilon \sim F_b$ 为条件来预测注入扰动。这种条件化将扩散目标转化为一个辅助的特征级正则化器：在不同时间步上进行准确的扰动估计，要求 F_b 保留结构一致的语义信息，同时对雾相关退化保持信息量。为了通过条件 U-Net 有效地传递这种退化感知引导， F_b 在两个不同的语义尺度上进行了空间对齐和融合。

在输入层， F_b 被双线性上采样到原始分辨率，经过 1×1 投影压缩，并与噪声残差 R_t 拼接。这种早期注入提供了像素对齐的全局上下文，使浅层卷积层能够区分空间雾分布与底层结构细节。

在网络更深处， F_b 被上采样到半分辨率尺度，并以加法方式整合到中间编码器表示中。这种次级融合在更高抽象层次上强化了语义先验，防止条件信号在特征经历逐步空间下采样时衰减。

与这种多尺度空间条件化相结合，扩散时间步 t 通过正弦嵌入和多层感知机进行编码，然后在 U-Net 层级结构中逐层相加。这种时间调制确保了所有层都能持续感知去噪轨迹上当前的扰动水平。形式上，辅助分支预测：

$$\hat{\epsilon} = \epsilon_{\theta}(R_t, t, \hat{F}_b) \quad (9)$$

where $\hat{\epsilon}$ denotes the predicted noise at timestep t , and ϵ is the ground-truth Gaussian noise added during the forward diffusion process. Here, ϵ_{θ} represents the conditional U-Net of the auxiliary branch. Since the residual $R = I_h - I_c$ explicitly represents the degradation introduced by haze, diffusion is performed directly in the degradation residual space. At each timestep, R_t is a perturbed version of R , and predicting the added noise provides the denoising supervision needed to estimate the underlying haze residual from different noise levels. Training across multiple timesteps therefore encourages the conditional U-Net to learn the statistical characteristics of haze residuals, including their degradation strength and spatial variation. Because noise prediction is conditioned on $\hat{F}_b$, the diffusion loss is backpropagated through ZI-PPH to the S-Encoder and LGCB modules. As a result, haze-dependent feature deviations that hinder consistent residual prediction across different noise levels are progressively penalized during training, encouraging the shared bottleneck to form a more structure-consistent and degradation-aware representation. This implicit regularization reduces the influence of haze-induced feature shifts without requiring an explicit feature-distribution alignment loss, while preserving structural and semantic information that is also beneficial to image reconstruction in the main decoder.

Notably, ZI-PPH introduces no inference-time overhead. During testing, the ZI-PPH branch is detached, and the network reconstructs the clear image via a single forward pass.

$$\hat{\epsilon} = \epsilon_{\theta}(R_t, t, \hat{F}_b) \quad (9)$$

其中 ϵ 表示时间步 t 处的预测噪声， ϵ 是前向扩散过程中添加的真实高斯噪声。这里， ϵ_{θ} 表示辅助分支的条件U-Net。由于残差 $R = I_h - I_c$ 显式地表示了雾霾引入的退化，扩散直接在退化残差空间中进行。在每个时间步， R_t 是 R 的扰动版本，预测添加的噪声提供了从不同噪声水平估计潜在雾霾残差所需的去噪监督。因此，跨多个时间步的训练促使条件U-Net学习雾霾残差的统计特征，包括其退化强度和空间变化。由于噪声预测以 F_b 为条件，扩散损失通过ZI-PPH反向传播到S-Encoder和LGCB模块。结果，在不同噪声水平下阻碍一致残差预测的雾霾相关特征偏差在训练过程中被逐步惩罚，鼓励共享瓶颈形成更加结构一致且具备退化感知的表示。这种隐式正则化在无需显式特征分布对齐损失的情况下，减少了雾霾引起的特征偏移的影响，同时保留了同样有利于主解码器中图像重建的结构与语义信息。

值得注意的是，ZI-PPH 不会引入任何推理时间开销。在测试过程中，ZI-PPH 分支被分离，网络通过单次前向传播重建清晰图像。

$$\widehat{I} = f_{\theta}(I_h, I_{hf}^h) \quad (10)$$

where $\widehat{I}$ denotes the final predicted clear image, and f_{θ} represents the deterministic feed-forward dehazing backbone parameterized by θ . This formulation bypasses all residual noising steps, conditional U-Net evaluations, and reverse diffusion chains. Thus, ZI-PPH transfers the representation-shaping benefits of diffusion supervision to the shared backbone during training, while maintaining the deterministic efficiency of a feed-forward architecture.

3.3. Training Objectives

The overall training objective of ZID-Net is a weighted combination of three terms:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_{contrast} + \lambda_3 \mathcal{L}_{diff} \quad (11)$$

where the empirically set hyperparameters are $\lambda_1 = 1.0$, $\lambda_2 = 0.1$, and $\lambda_3 = 0.35$. The reconstruction term employs a standard pixel-wise l_1 loss to constrain the fidelity between the predicted clear image $\widehat{I}$ and the ground truth I_c .

$$\mathcal{L}_1 = \|\widehat{I} - I_c\|_1 \quad (12)$$

To enhance perceptual quality, we utilize a contrastive loss computed in the VGG-19 feature space. It pulls the restored image closer to the clean target (positive) while pushing it away from both the hazy input and a degradation-focused noisy residual (negatives). Let $\phi_l(\cdot)$ denote the extracted features from the l -th layer of a pre-trained VGG-19 network [30], with corresponding weights ω_l . Defining the clipped noisy residual image as $\widetilde{I}_h^t = \text{clip}(I_c + R_t, 0, 1)$, the contrastive loss is formulated as:

$$\widehat{I} = f_{\theta}(I_h, I_{hf}^h) \quad (10)$$

其中 I 表示最终预测的清晰图像, f_{θ} 表示由 θ 参数化的确定性前馈去雾骨干网络。该公式绕过了所有残差加噪步骤、条件U-Net评估和逆向扩散链。因此, ZI-PPH 在训练期间将扩散监督的表示塑造优势传递给共享骨干网络, 同时保持了前馈架构的确定性效率。

3.3. 训练目标

ZID-Net 的总体训练目标是三个项的加权组合:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_{contrast} + \lambda_3 \mathcal{L}_{diff} \quad (11)$$

其中经验设定的超参数为 $\lambda_1 = 1.0$ 、 $\lambda_2 = 0.1$ 和 $\lambda_3 = 0.35$ 。重建项采用标准的逐像素 l_1 损失来约束预测清晰图像 I 与真值 I_c 之间的保真度。

$$\mathcal{L}_1 = \|\widehat{I} - I_c\|_1 \quad (12)$$

为了提升感知质量, 我们在VGG-19特征空间中计算对比损失。该损失将恢复图像拉近清晰目标(正样本), 同时将其推离有雾输入和面向退化的噪声残差(负样本)。令 $\phi_l(\cdot)$ 表示从预训练VGG-19网络 [30], 的第 l 层提取的特征, 对应权重为 ω_l 。将截断噪声残差图像定义为 $I_h^t = \text{clip}(I_c + R_t, 0, 1)$, 对比损失表示为:

$$\mathcal{L}_{contrast} = \sum_l \omega_l \frac{\|\phi_l(\widehat{I}) - \phi_l(I_c)\|_1}{\|\phi_l(\widehat{I}) - \phi_l(I_h)\|_1 + \|\phi_l(\widehat{I}) - \phi_l(\widetilde{I}_h^t)\|_1 + \varepsilon} \quad (13)$$

where $\varepsilon = 10^{-7}$ prevents numerical instability. Here, I_h acts as an image-domain negative, while $\widetilde{I}_h^t$ serves as a degradation-focused negative explicitly tied to the diffusion perturbation.

Finally, the diffusion supervision term penalizes the l_1 distance between the predicted noise $\widehat{\epsilon}$ and the sampled Gaussian noise ϵ :

$$\mathcal{L}_{diff} = \|\widehat{\epsilon} - \epsilon\|_1 \quad (14)$$

Together, these three losses tightly couple pixel-level reconstruction, perceptual discrimination, and generative degradation modeling into a single, cohesive optimization framework.

4. Experiments

In this section, we comprehensively evaluate the effectiveness and generalization of our ZID-Net. Section 4.1 describes the experimental setup, including datasets, evaluation metrics, and baselines. Then, we assess our model on synthetic haze in Section 4.2, real-world conditions in Section 4.3, and remote sensing imagery in Section 4.4. Section 4.5 analyzes the computational efficiency of our deterministic architecture, and Section 4.6 presents ablation studies to validate the core components.

4.1. Experimental Setup

Implementation details. All experiments are implemented in PyTorch and conducted on a single NVIDIA GeForce RTX 5060 Ti GPU with 16 GB of VRAM. The network is trained using the Adam optimizer with default momentum parameters and a batch size of 8. The initial learning rate is set to 0.0001 and gradually decays via cosine annealing. To augment the training data,

$$\mathcal{L}_{contrast} = \sum_l \omega_l \frac{\|\phi_l(\widehat{I}) - \phi_l(I_c)\|_1}{\|\phi_l(\widehat{I}) - \phi_l(I_h)\|_1 + \|\phi_l(\widehat{I}) - \phi_l(\widetilde{I}_h^t)\|_1 + \varepsilon} \quad (13)$$

其中 $\varepsilon = 10^{-7}$ 用于防止数值不稳定。这里, I_h 作为~图像域负样本, 而 I_h^t 则作为明确与扩散扰动相关联的退化聚焦负样本。

最后, 扩散监督项对 l_1 ~预测噪声 ϵ 与采样高斯噪声 ϵ 之间的 l_1 距离进行惩罚:

$$\mathcal{L}_{diff} = \|\widehat{\epsilon} - \epsilon\|_1 \quad (14)$$

这三个损失共同将像素级重建、感知判别和生成式退化建模紧密耦合到一个统一的优化框架中。

4. 实验

在本节中, 我们全面评估 ZID-Net 的有效性和泛化能力。第 4.1 节描述了实验设置, 包括数据集、评估指标和基线。随后, 我们在第 4.2 节评估模型在合成雾霾上的表现, 第 4.3 节评估真实世界条件下的表现, 第 4.4 节评估遥感图像上的表现。第 4.5 节分析了确定性架构的计算效率, 第 4.6 节通过消融研究验证核心组件的有效性。

4.1. 实验设置

实现细节。所有实验均在PyTorch中实现, 并在配备16 GB显存的单张NVIDIA GeForce RTX 5060 Ti GPU上进行。网络使用具有默认动量参数的Adam优化器进行训练, 批大小为8。初始学习率设置为0.0001, 并通过余弦退火逐渐衰减。为增强训练数据,

we randomly crop the images into 256×256 spatial patches, followed by random rotations, flips, and mild scaling. For the ZI-PPH branch, the diffusion prior employs $T = 1000$ forward steps. The contrastive loss is computed using multi-scale features extracted from a pre-trained VGG-19 network [30].

Datasets. We evaluate our model across synthetic, real-world, and remote sensing scenarios. For synthetic haze, we use the RESIDE [31] benchmark, training on ITS and OTS, and testing on the corresponding SOTS-Indoor and SOTS-Outdoor sets. To evaluate robustness across diverse real-world scenes, we establish the Real-World Thin setting by merging NH-HAZE [32], I-HAZE [33], and O-HAZE [34]. This provides 155 pairs covering indoor and outdoor scenes with homogeneous and non-homogeneous haze, which are partitioned into 145 training and 10 testing pairs. For extreme conditions, the Real-World Dense setting utilizes the Dense-Haze [35] dataset with 50 training and 5 testing pairs to assess structural recovery under severe physical degradation. Finally, for remote sensing, we employ the StateHaze1k [36] dataset, which comprises Thin, Moderate, and Thick subsets, each providing 320 training and 45 testing pairs.

Evaluation metrics. We quantitatively assess restoration quality using Peak Signal-to-Noise Ratio (PSNR) [37] and Structural Similarity Index (SSIM) [38]. To evaluate the computational efficiency of our feed-forward architecture, we report the trainable parameters, GFLOPs, and average inference runtime in milliseconds. Both FLOPs and runtime are measured at 256×256 and 512×512 resolutions to demonstrate the linear scaling complexity.

Baselines. We compare ZID-Net against twelve state-of-the-art dehazing methods: the physical prior-based DCP [8]; CNN-based models including FFA

我们将图像随机裁剪为 256×256 空间块，随后进行随机旋转、翻转和轻度缩放。对于ZI-PPH分支，扩散先验采用 $T = 1000$ 个前向步骤。对比损失使用从预训练VGG-19网络提取的多尺度特征计算 [30]。

数据集。我们在合成、真实世界和遥感场景下评估了我们的模型。对于合成雾霾，我们使用RESIDE [31] 基准，在ITS和OTS上进行训练，并在相应的SOTS-Indoor和SOTS-Outdoor集上进行测试。为了评估在各种真实世界场景中的鲁棒性，我们通过合并NH-HAZE [32], I-HAZE[33], 和 O-HAZE [34]建立了真实世界薄雾设置。这提供了155对覆盖室内外场景的均匀雾霾和非均匀雾霾图像，划分为145对训练图像和10对测试图像。对于极端条件，真实世界浓雾设置利用Dense-Haze [35] 数据集，包含 50对训练图像和5对测试图像，以评估严重物理退化下的结构恢复。最后，对于遥感，我们采用StateHaze1k [36] 数据集，该数据集包含薄雾、中等雾和浓雾子集，每个子集提供320对训练图像和45对测试图像。

评估指标。我们使用峰值信噪比（PSNR）[37] 和结构相似性指数（SSIM）[38]对复原质量进行定量评估。为了评估我们前馈架构的计算效率，我们报告了可训练参数、GFLOPs 以及以毫秒为单位的平均推理运行时间。FLOPs 和运行时间均在 256×256 和 512×512分辨率下测量，以展示线性扩展复杂度。

基线。我们将 ZID-Net 与十二种最先进的去雾方法进行比较：基于物理先验的 DCP [8]；基于 CNN 的模型，包括 FFA

[12], SGID [39], C2PNet [13], SANet [40], DCMNet [14], DEA-Net [15], SGDN [16], and TUR [41]; Transformer-based architectures including DehazeFormer [21] and MB-TaylorFormerV2 [23]; and the diffusion-based *Learning Hazing to Dehazing* [29]. For qualitative visual comparisons, we select a subset of these baselines: DCP [8], C2PNet [13], DCMNet [14], SGDN [16], DehazeFormer [21], MB-TaylorFormerV2 [23], and *Learning Hazing to Dehazing* [29].

4.2. Evaluation on Synthetic Datasets

We first assess the restoration capabilities on the synthetic RESIDE benchmark. Quantitatively, as detailed in Table 1, ZID-Net establishes state-of-the-art performance on the synthetic RESIDE benchmark. On the SOTS-Indoor dataset, our model reaches a PSNR of 42.77 dB and an SSIM of 0.997. This advantage continues on the SOTS-Outdoor dataset, where ZID-Net exceeds the second-best method by 0.72 dB, achieving a PSNR of 38.73 dB and an SSIM of 0.993. This margin confirms that our frequency-spatial decoupled backbone effectively preserves structural and chromatic details under synthetic degradation.

Table 1

Quantitative comparison results with representative dehazing methods on SOTS-Indoor and SOTS-Outdoor. For each metric, the best results are highlighted in red and second-best in blue.

Method	Venue & Year	SOTS-Indoor		SOTS-Outdoor		Average	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
DCP	TPAMI 2010	19.50	0.859	16.98	0.858	18.24	0.859
FFA	AAAI 2020	36.39	0.989	33.57	0.984	34.98	0.986
SGID	TIP 2022	38.52	0.991	34.36	0.983	36.44	0.987
C2PNet	CVPR 2023	42.56	0.995	36.68	0.990	39.62	0.993
Dehaze Former-B	TIP 2023	37.84	0.994	34.95	0.984	36.40	0.989
SANet	IJCAI 2023	40.40	0.996	38.01	0.995	39.21	0.996
DCMNet	CVPR 2024	42.18	0.997	36.56	0.993	39.37	0.995

[12], SGID [39], C2PNet [13], SANet [40], DCMNet [14], DEA-Net [15],SGDN [16], and TUR [41]; 基于 Transformer 的架构, 包括 DehazeFormer [21] 和 MB-TaylorFormerV2 [23]; 以及基于扩散的*Learning Hazing to Dehazing* [29]。对于定性视觉比较, 我们选择了这些基线的子集: DCP [8], C2PNet [13], DCMNet [14],SGDN [16], DehazeFormer [21],MB-TaylorFormerV2 [23], 和 *LearningHazingto Dehazing* [29]。

4.2. 合成数据集上的评估

我们首先在合成RESIDE基准上评估复原能力。从定量角度看, 如表1所述, ZID-Net在合成RESIDE基准上建立了最先进的性能。在SOTS-Indoor数据集上, 我们的模型达到了42.77 dB的峰值信噪比和0.997的结构相似性。这一优势在SOTS-Outdoor数据集上得以延续, ZID-Net以0.72 dB的优势超过第二名方法, 达到了38.73 dB的峰值信噪比和0.993的结构相似性。这一差距证实了我们的频域-空间解耦主干网络在合成退化下有效保持了结构细节和色彩细节。

表1与代表性去雾方法在SOTS-Indoor和SOTS-Outdoor上的定量比较结果。对于每项指标, 最佳结果以红色标出, 次佳结果以蓝色标出。

方法	发表来源与 Year	SOTS-Indoor		SOTS-Outdoor		Average	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
DCP	TPAMI 2010	19.50	0.859	16.98	0.858	18.24	0.859
FFA	AAAI 2020	36.39	0.989	33.57	0.984	34.98	0.986
SGID	TIP 2022	38.52	0.991	34.36	0.983	36.44	0.987
C2PNet	CVPR 2023	42.56	0.995	36.68	0.990	39.62	0.993
Dehaze Former-B	TIP 2023	37.84	0.994	34.95	0.984	36.40	0.989
SANet	IJCAI 2023	40.40	0.996	38.01	0.995	39.21	0.996
DCMNet	CVPR 2024	42.18	0.997	36.56	0.993	39.37	0.995

DEA-Net	TIP 2024	41.31	0.995	36.59	0.990	38.95	0.993
MB-TaylorFormerV2-B	TPAMI 2025	41.00	0.993	37.81	0.991	39.41	0.992
TUR	AAAI 2025	31.17	0.976	28.75	0.972	29.96	0.974
SGDN	AAAI 2025	35.56	0.985	31.92	0.981	33.74	0.983
<i>Learning Hazing to Dehazing</i>	CVPR 2025	14.77	0.540	17.31	0.588	16.04	0.564
Ours	-	42.77	0.997	38.73	0.993	40.75	0.995

Consistent with the quantitative metrics, the qualitative evaluations validate our architectural advantages. As visualized in Figs. 4 and 5, the traditional DCP frequently exhibits color shifts and illumination imbalances. In contrast, deterministic deep learning methods demonstrate superior restoration consistency, effectively removing synthetic haze. However, a closer inspection of the magnified regions reveals that the diffusion-based *Learning Hazing to Dehazing* [29] approach exhibits artifacts stemming from the inherent stochasticity of its sampling process. ZID-Net avoids these generative instabilities. By employing a deterministic inference architecture, our method provides stable dehazing performance comparable to advanced baselines while maintaining higher color fidelity and sharper structural boundaries.

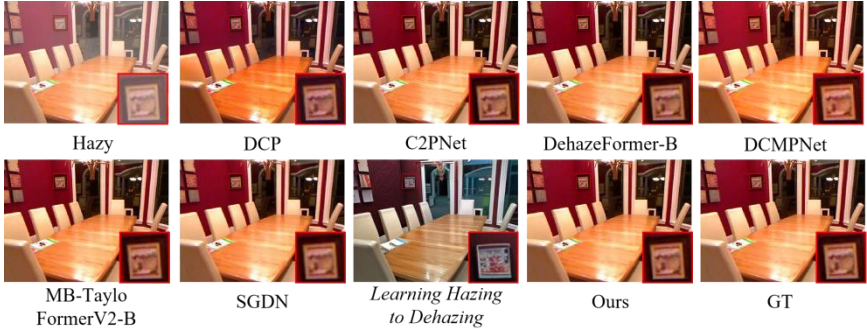


Fig. 4. Comparison results of the proposed method with each method on SOTS-Indoor dataset.

DEA-Net	TIP 2024	41.31	0.995	36.59	0.990	38.95	0.993
MB-TaylorFormerV2-B	TPAMI 2025	41.00	0.993	37.81	0.991	39.41	0.992
TUR	AAAI 2025	31.17	0.976	28.75	0.972	29.96	0.974
SGDN	AAAI 2025	35.56	0.985	31.92	0.981	33.74	0.983
<i>Learning Hazing to Dehazing</i>	CVPR 2025	14.77	0.540	17.31	0.588	16.04	0.564
Ours	-	42.77	0.997	38.73	0.993	40.75	0.995

与定量指标一致，定性评估验证了我们的架构优势。如图4和图5所展示，传统DCP经常出现色彩偏移和光照不均衡。相比之下，确定性深度学习方法表现出更优的复原一致性，能够有效去除合成雾霾。然而，对放大区域的仔细检查显示，基于扩散的*Learning Hazing to Dehazing* [29] 方法由于采样过程的固有随机性而表现出伪影。ZID-Net避免了这些生成不稳定性。通过采用确定性推理架构，我们的方法提供了与先进基线相当的稳定去雾性能，同时保持更高的色彩保真度和更清晰的结构边界。

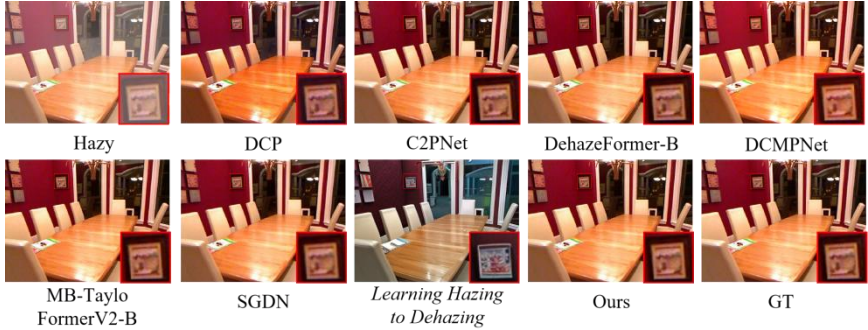


图4. 比较 n 结果：所提出的方法与各方法在 SOTS-I 上的对比 ndoor 数据集。

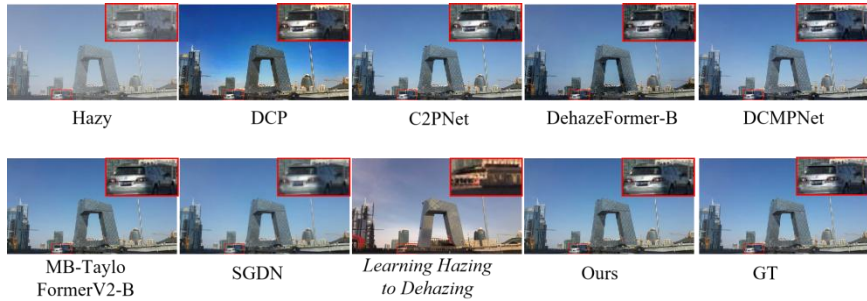


Fig. 5. Comparison results of the proposed method with each method on SOTS-Outdoor dataset.

4.3. Evaluation on Real-World Datasets

To verify generalization in practical scenarios, Table 2 reports the quantitative results on the Real-World Thin and Dense datasets. Standard diffusion models generally underperform on pixel-aligned metrics like PSNR and SSIM because their generative formulation favors perceptual quality and distribution alignment over deterministic spatial fidelity. ZID-Net circumvents this limitation by utilizing a deterministic inference architecture guided by diffusion priors. On the Real-World Thin dataset, our method outperforms the second-best approach by 1.37 dB in PSNR and 0.044 in SSIM. For the Real-World Dense dataset, which contains scenes with severe haze and substantial visibility obstruction, ZID-Net achieves a 0.97 dB PSNR improvement over the previous state-of-the-art, together with an SSIM of 0.601. This result demonstrates its ability to maintain effective structural perception and pixel-level restoration even under heavily obscured conditions.

Table 2

Quantitative comparison results with excellent methods on Real-World Thin and Real-World Dense. For each metric, the best results are highlighted in red and second-best in blue.

Method	Venue & Year	Real-World Thin		Real-World Dense		Average	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
DCP	TPAMI 2010	14.46	0.599	13.43	0.465	13.95	0.532
FFA	AAAI 2020	15.86	0.637	18.80	0.577	17.33	0.607
SGID	TIP 2022	17.47	0.681	15.57	0.558	16.52	0.620

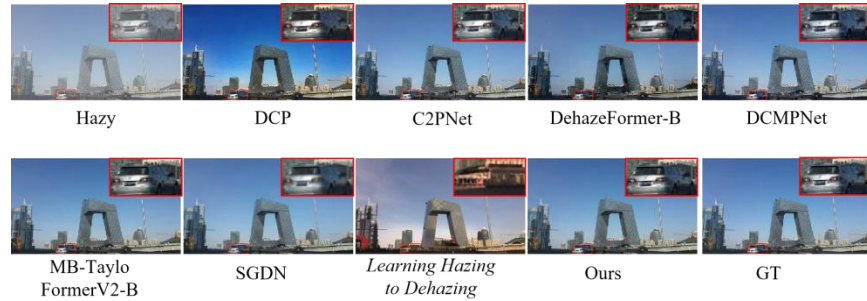


图 5. 所提方法与各方法在 SOTS-Outdoor 数据集上的对比结果。

4.3. 在真实世界数据集上的评估

为了验证在实际场景中的泛化能力，表2报告了在真实世界薄雾和浓雾数据集上的定量结果。标准扩散模型在PSNR和SSIM等像素对齐指标上通常表现不佳，因为其生成式公式更倾向于感知质量和分布对齐，而非确定性的空间保真度。ZID-Net通过利用由扩散先验引导的确定性推理架构，规避了这一局限。在真实世界薄雾数据集上，我们的方法在PSNR上比第二名方法高出1.37 dB，在SSIM上高出0.044。对于包含严重雾霾和显著能见度遮挡场景的真实世界浓雾数据集，ZID-Net相比此前的先进方法实现了0.97 dB的PSNR提升，同时SSIM达到0.601。这一结果表明，即使在严重遮挡条件下，它也能保持有效的结构感知和像素级恢复能力。

表 2 与优秀方法在真实世界薄雾和真实世界浓雾上的定量比较结果。对于每项指标，最佳结果以红色标注，次佳结果以蓝色标注。

方法	发表来源 & Year	真实世界薄雾		Real-World Dense		Average	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	峰值信噪比↑	结构相似性↑
DCP	TPAMI 2010	14.46	0.599	13.43	0.465	13.95	0.532
FFA	AAAI 2020	15.86	0.637	18.80	0.577	17.33	0.607
SGID	TIP 2022	17.47	0.681	15.57	0.558	16.52	0.620

C2PNet	CVPR 2023	18.83	0.708	16.86	0.558	17.85	0.633
Dehaze Former-B	TIP 2023	16.65	0.692	18.49	0.602	17.57	0.647
SANet	IJCAI 2023	17.42	0.724	18.11	0.595	17.77	0.660
DCMPNet	CVPR 2024	15.31	0.653	13.32	0.149	14.32	0.401
DEA-Net	TIP 2024	18.53	0.719	19.52	0.533	19.03	0.626
MB-TaylorFormerV2-B	TPAMI 2025	19.37	0.702	19.40	0.593	19.39	0.648
TUR	AAAI 2025	19.46	0.721	12.65	0.499	16.06	0.610
SGDN	AAAI 2025	20.46	0.732	19.90	0.623	20.18	0.678
<i>Learning Hazing to Dehazing</i>	CVPR 2025	15.02	0.393	13.21	0.431	14.12	0.412
Ours	-	21.83	0.776	20.87	0.601	21.35	0.689

The visual advantages of ZID-Net are evident under authentic degradation conditions. A closer inspection of the magnified floral regions in Fig. 6 reveals that convolutional networks such as C2PNet [13], DCMPNet [14], and SGDN [16] exhibit overexposure, which leads to the loss of fine-grained petal textures and structural details. In contrast, ZID-Net maintains balanced illumination and preserves these high-frequency components. Furthermore, as shown in Fig. 7, competing methods exhibit chromatic distortion and fail to recover intrinsic colors under dense haze. In contrast, the scene-level global perception provided by LGCBs enables ZID-Net to maintain balanced local exposure, preventing color shifts during real-world image restoration.

C2PNet	CVPR 2023	18.83	0.708	16.86	0.558	17.85	0.633
Dehaze Former-B	TIP 2023	16.65	0.692	18.49	0.602	17.57	0.647
SANet	IJCAI 2023	17.42	0.724	18.11	0.595	17.77	0.660
DCMPNet	CVPR 2024	15.31	0.653	13.32	0.149	14.32	0.401
DEA-Net	TIP 2024	18.53	0.719	19.52	0.533	19.03	0.626
MB-TaylorFormerV2-B	TPAMI 2025	19.37	0.702	19.40	0.593	19.39	0.648
TUR	AAAI 2025	19.46	0.721	12.65	0.499	16.06	0.610
SGDN	AAAI 2025	20.46	0.732	19.90	0.623	20.18	0.678
<i>Learning Hazing to Dehazing</i>	CVPR 2025	15.02	0.393	13.21	0.431	14.12	0.412
Ours	-	21.83	0.776	20.87	0.601	21.35	0.689

在真实退化条件下，ZID-Net的视觉优势十分明显。对图6中放大的花卉区域进行仔细检查可以发现，C2PNet [13], DCMPNet [14], 和SGDN [16] 等卷积网络会出现过曝，从而导致细粒度花瓣纹理和结构细节的丢失。相比之下，ZID-Net保持了均衡的光照并保留了这些高频分量。此外，如图7所示，竞争方法在浓雾下会出现色彩失真，无法恢复固有颜色。相比之下，LGCBs提供的场景级全局感知使ZID-Net能够保持均衡的局部曝光，从而防止真实世界图像复原过程中的色彩偏移。

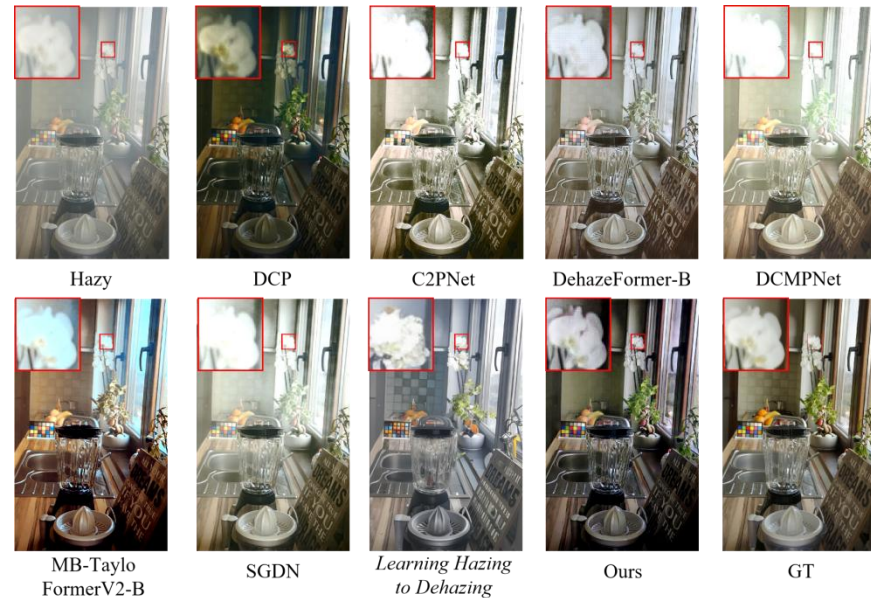


Fig. 6. Comparison results of the proposed method with each method on Real-World Thin dataset.

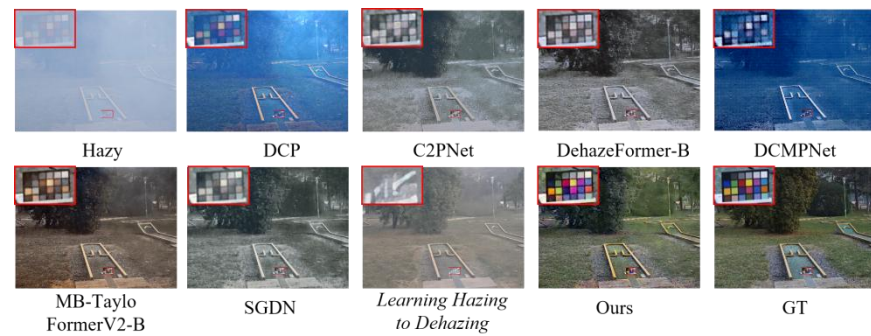


Fig. 7. Comparison results of the proposed method with each method on Real-World Dense dataset.

4.4. Evaluation on Remote Sensing Imagery

We further extend the evaluation to the remote sensing domain using the StateHaze1k dataset. As reported in Table 3, our prior-guided deterministic architecture establishes a consistent advantage on the StateHaze1k dataset across varying density levels. By utilizing the diffusion prior to inject severity-aware structural guidance into the deterministic backbone, ZID-Net achieves PSNR improvements of 1.16 dB and 1.36 dB over the second-best approaches under Thin and Moderate conditions. The architectural benefit becomes particularly evident under Thick haze, where severe spatial degradation

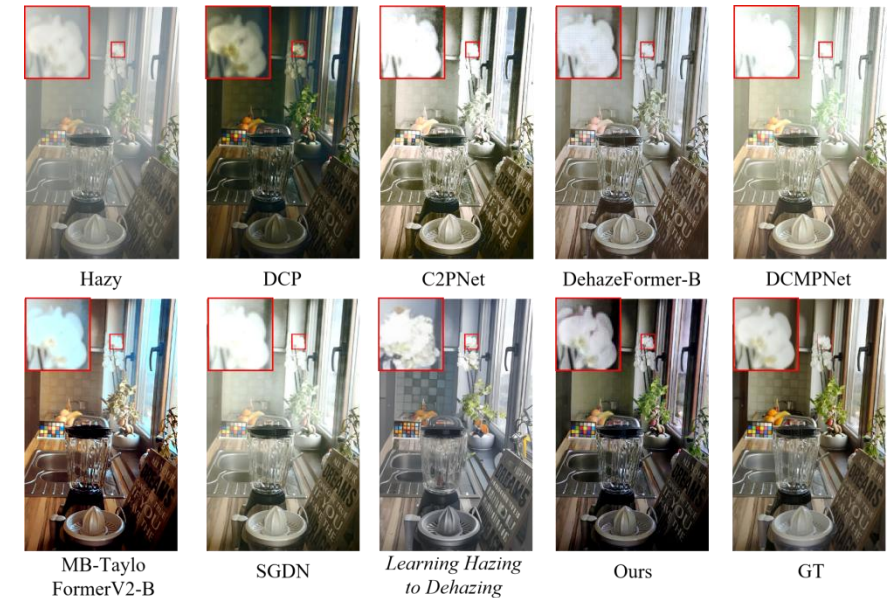


Fig. 6. Comparison results of the proposed method with each method on Real-World Thin dataset.

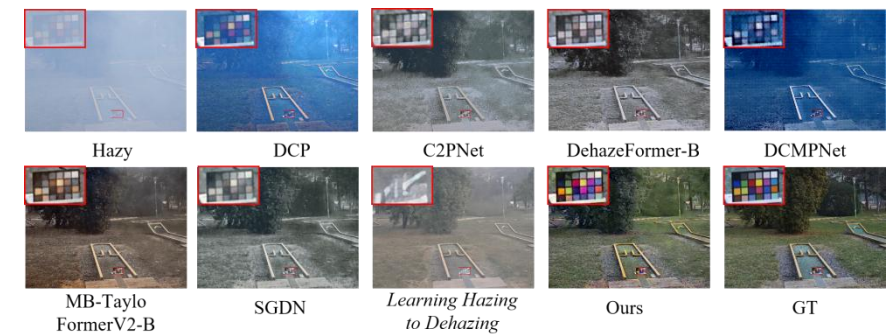


Fig. 7. Comparison results of the proposed method with each method on Real-World Dense dataset.

4.4. Evaluation on Remote Sensing Imagery

我们进一步将评估扩展到遥感领域，使用StateHaze1k数据集。如表3所示，我们提出的先验引导的确定性架构在StateHaze1k数据集上，跨不同密度等级均建立了稳定的优势。通过利用扩散先验将严重度感知的结构引导注入确定性骨干网络，ZID-Net在薄雾和中等雾条件下，相比次优方法分别实现了1.16 dB和1.36 dB的峰值信噪比提升。该架构优势在浓雾条件下尤为明显，此时严重的空间退化

typically limits conventional feature extraction. In this highly degraded scenario, the diffusion prior regularizes the feature space, allowing our method to surpass the next best performance by 5.57 dB in PSNR and 0.061 in SSIM. Overall, ZID-Net achieves the highest average PSNR of 25.29 dB and an SSIM of 0.915, leading the closest baseline by an average of 3.06 dB in PSNR and 0.032 in SSIM. These quantitative gains confirm that integrating generative priors into a feed-forward network robustly mitigates complex remote sensing degradation.

Table 3

Quantitative evaluation results on the remote sensing dataset StateHaze1k under different haze densities (Thin, Moderate, and Thick). For each metric, the best results are highlighted in red and second-best in blue.

Method	Venue & Year	Thin		Moderate		Thick		Average	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
DCP	TPAMI 2010	20.61	0.886	21.55	0.919	16.73	0.773	19.63	0.859
FFA	AAAI 2020	21.29	0.895	24.23	0.926	12.14	0.648	19.22	0.823
SGID	TIP 2022	21.69	0.906	23.68	0.940	12.29	0.692	19.22	0.846
C2PNet	CVPR 2023	22.66	0.909	25.76	0.940	14.66	0.761	21.02	0.870
Dehaze Former-B	TIP 2023	24.10	0.916	26.26	0.940	15.83	0.751	22.06	0.869
SANet	IJCAI 2023	22.69	0.907	24.86	0.940	17.34	0.766	21.63	0.871
DCMPNet	CVPR 2024	22.87	0.905	24.09	0.912	14.83	0.728	20.60	0.849
DEA-Net	TIP 2024	23.73	0.903	25.72	0.923	17.24	0.789	22.23	0.872
MB-TaylorFormerV2-B	TPAMI 2025	21.12	0.898	24.62	0.944	16.52	0.808	20.75	0.883
TUR	AAAI 2025	23.28	0.913	26.35	0.939	14.55	0.771	21.39	0.874
SGDN	AAAI 2025	22.67	0.907	25.69	0.940	13.32	0.735	20.56	0.861
Learning Hazing to Dehazing	CVPR 2025	13.79	0.217	15.55	0.214	13.49	0.196	14.28	0.209
Ours	-	25.26	0.928	27.71	0.947	22.91	0.869	25.29	0.915

The visual comparisons directly corroborate these quantitative gains. Figs. 8, 9 and 10 demonstrate the results across thin, moderate, and thick haze conditions, respectively. As the haze density increases, baseline methods such as DehazeFormer [21] and MB-TaylorFormerV2 [23] degrade significantly. They

通常会限制传统特征提取。在这种高度退化的场景中，扩散先验对特征空间进行正则化，使我们的方法在峰值信噪比上超越次优表现5.57 dB，在结构相似性上超越0.061。总体而言，ZID-Net取得了最高的平均峰值信噪比25.29 dB和0.915的结构相似性，在平均峰值信噪比上领先最接近的基线3.06 dB，在结构相似性上领先0.032。这些定量增益证实，将生成先验整合到前馈网络中，能够稳健地缓解复杂的遥感图像退化。

Table 3

Quantitative evaluation results on the remote sensing dataset StateHaze1k under different haze densities (Thin, Moderate, and Thick). For each metric, the best results are highlighted in red and second-best in blue.

Method	Venue & Year	Thin		Moderate		Thick		Average	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
DCP	TPAMI 2010	20.61	0.886	21.55	0.919	16.73	0.773	19.63	0.859
FFA	AAAI 2020	21.29	0.895	24.23	0.926	12.14	0.648	19.22	0.823
SGID	TIP 2022	21.69	0.906	23.68	0.940	12.29	0.692	19.22	0.846
C2PNet	CVPR 2023	22.66	0.909	25.76	0.940	14.66	0.761	21.02	0.870
Dehaze Former-B	TIP 2023	24.10	0.916	26.26	0.940	15.83	0.751	22.06	0.869
SANet	IJCAI 2023	22.69	0.907	24.86	0.940	17.34	0.766	21.63	0.871
DCMPNet	CVPR 2024	22.87	0.905	24.09	0.912	14.83	0.728	20.60	0.849
DEA-Net	TIP 2024	23.73	0.903	25.72	0.923	17.24	0.789	22.23	0.872
MB-TaylorFormerV2-B	TPAMI 2025	21.12	0.898	24.62	0.944	16.52	0.808	20.75	0.883
TUR	AAAI 2025	23.28	0.913	26.35	0.939	14.55	0.771	21.39	0.874
SGDN	AAAI 2025	22.67	0.907	25.69	0.940	13.32	0.735	20.56	0.861
学习加雾至去雾	CVPR 2025	13.79	0.217	15.55	0.214	13.49	0.196	14.28	0.209
Ours	-	25.26	0.928	27.71	0.947	22.91	0.869	25.29	0.915

视觉对比直接印证了这些定量增益。图8、图9和图10分别展示了薄雾、中等雾和浓雾条件下的结果。随着雾浓度增加，DehazeFormer [21] 和 MB-TaylorFormerV2 [23] 等基线方法性能显著下降。它们

fail to fully remove the haze layer, leaving residual artifacts that obscure geographical features. The magnified regions in Fig. 10 further reveal that DehazeFormer [21] introduces severe color distortion under thick, non-homogeneous haze. In contrast, ZID-Net preserves sharp edges and structural details across different haze densities, including regions with severe local obscuration, and reliably reconstructs surface textures under highly restricted visibility. This robustness highlights the advantage of the dual-stream architecture. By using an independent high-frequency stream, the network can infer and recover structural information that is weakened or partially hidden by dense, spatially varying haze.

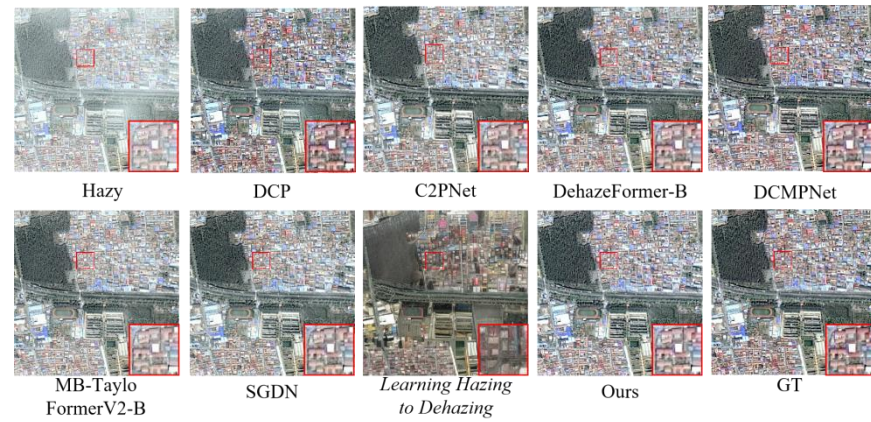


Fig. 8. Comparison results of the proposed method with each method on StateHaze1k Thin dataset.

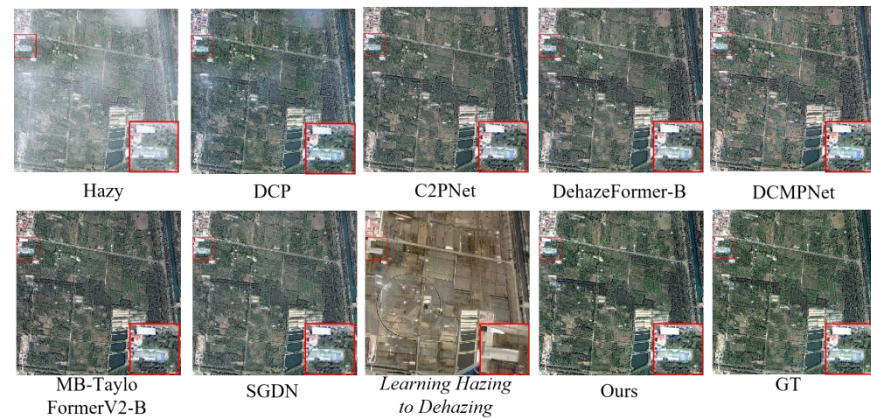


Fig. 9. Comparison results of the proposed method with each method on StateHaze1k Moderate dataset.

未能完全去除雾霾层，残留的伪影遮挡了地理特征。图10中的放大区域进一步揭示，DehazeFormer[21] 在浓重的非均匀雾霾下会产生严重的色彩失真。相比之下，ZID-Net在不同雾浓度下均能保持锐利边缘和结构细节，包括在严重局部遮挡的区域，并能在能见度极低的情况下可靠地重建表面纹理。这种鲁棒性凸显了双流架构的优势。通过使用独立的高频流，网络可以推断并恢复被浓密、空间变化的雾霾削弱或部分遮蔽的结构信息。

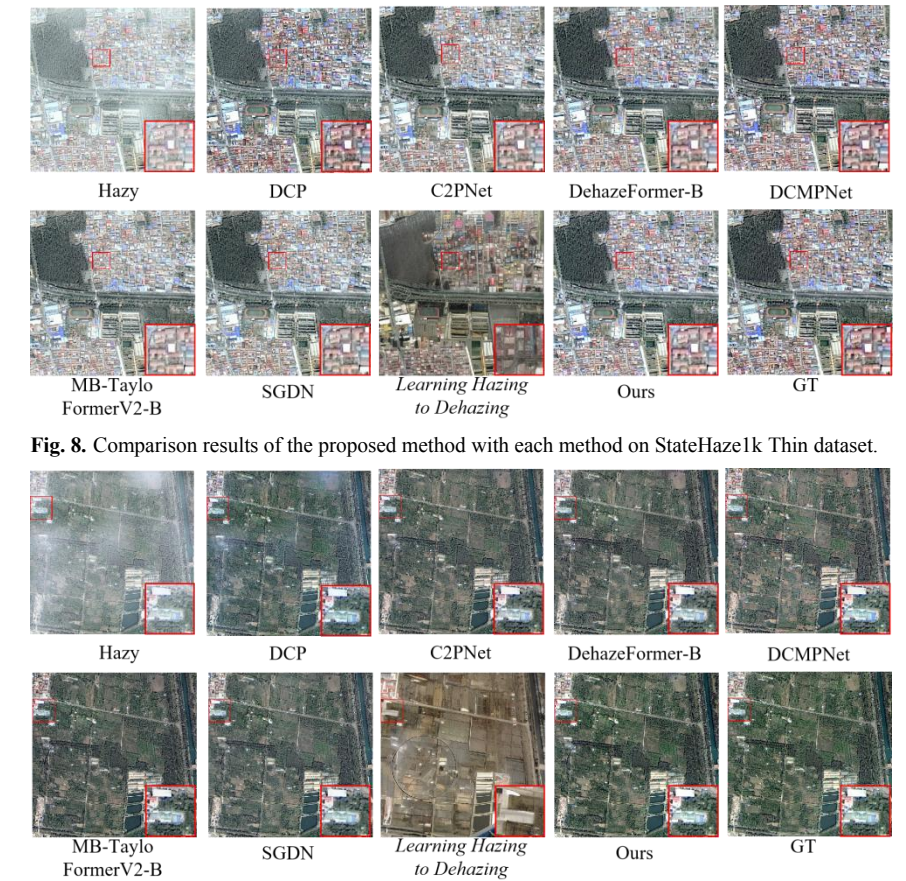


Fig. 8. Comparison results of the proposed method with each method on StateHaze1k Thin dataset.

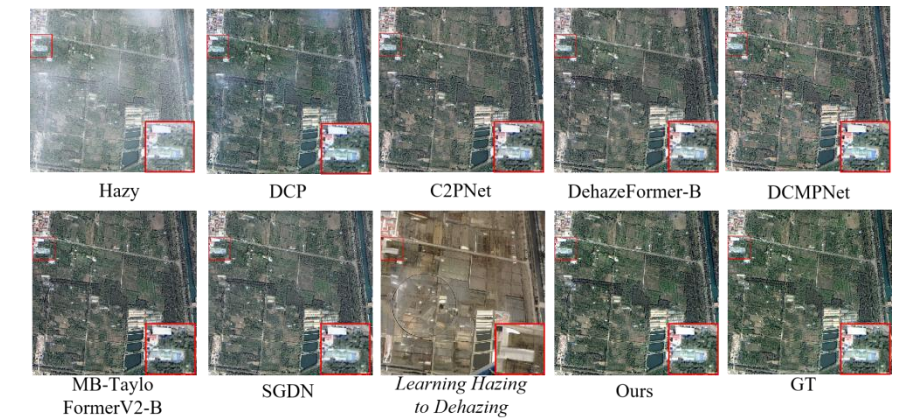


Fig. 9. Comparison results of the proposed method with each method on StateHaze1k Moderate dataset.

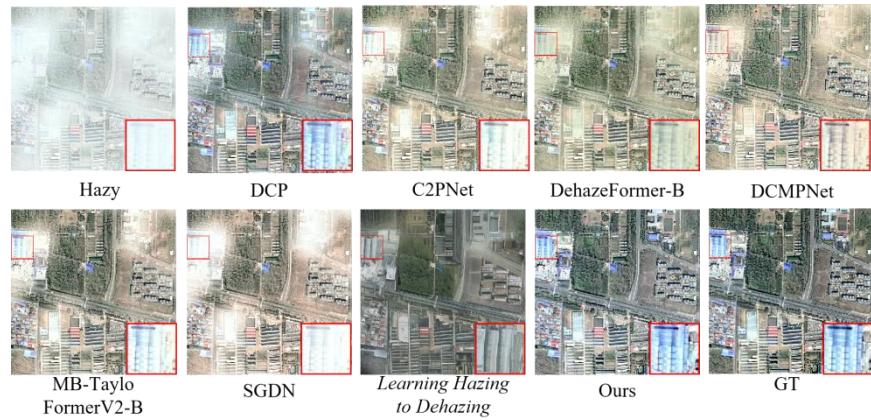


Fig. 10. Comparison results of the proposed method with each method on StateHaze1k Thick dataset.

4.5. Computational Efficiency Analysis

To evaluate practical applicability, we analyze the computational efficiency across 256×256 and 512×512 spatial resolutions, as detailed in Tables 4 and 5. Notably, the 512×512 evaluation aligns with the native resolution of the StateHaze1k dataset, directly reflecting the processing demands of remote sensing imagery. While ZID-Net maintains a moderate parameter count of 17.51 M to support its dual-stream feature extraction, it remains computationally efficient, requiring only 17.91 GMac and 71.64 GMac at the respective resolutions. The primary architectural advantage, however, emerges in inference latency. Because the diffusion prior module is detached during testing, ZID-Net achieves processing speeds of 5.14 ms and 19.35 ms, ranking as the fastest among all evaluated methods. This efficiency becomes particularly evident when compared to the standard diffusion-based method, *Learning Hazing to Dehazing* [29], which demands 9540.21 ms and 19870.79 ms per image due to iterative reverse sampling. Ultimately, our zero-inference deterministic design overcomes the inherent latency bottlenecks of diffusion models, enabling real-time inference without compromising restoration fidelity.

Table 4

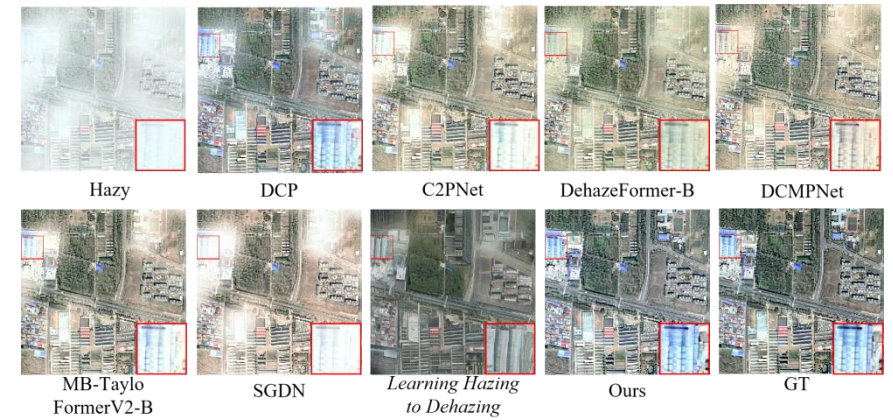


Fig. 10. Comparison results of the proposed method with each method on StateHaze1k Thick dataset.

4.5. Computational Efficiency Analysis

为评估实际适用性，我们分析了 256×256 和 512×512 空间分辨率下的计算效率，详见表4和表5。值得注意的是， 512×512 评估与 StateHaze1k数据集的原生分辨率一致，直接反映了遥感图像的处理需求。尽管ZID-Net为支持双流特征提取而保持了17.51 M的中等参数量，但其计算效率仍然较高，在两种分辨率下分别仅需17.91 GMac和71.64 GMac。然而，主要的架构优势体现在推理延迟上。由于扩散先验模块在测试期间被移除，ZID-Net的处理速度分别达到5.14 ms和19.35 ms，在所有评估方法中排名最快。与标准的基于扩散方法*Learning Hazing to Dehazing* [29]，相比，这一效率尤为突出，后者由于迭代逆向采样，每张图像需要9540.21 ms和19870.79 ms。最终，我们的零推理确定性设计克服了扩散模型固有的延迟瓶颈，在不牺牲恢复保真度的情况下实现了实时推理。

Table 4

Comparison of computational efficiency among representative dehazing models evaluated on 256 × 256 images.												
Method	FFA	SGID	C2PNe t	Dehaze Former- B	SAN et	DCMPNet	DEA-Ne t	MB-Taylor FormerV2-B	TUR	SGD N	<i>Learning Hazing to Dehazing</i>	Ours
Params (M)	4.46	13.87	7.17	2.52	3.81	17.36	3.65	2.63	2.60	11.08	1228.89	17.51
FLOPs (GMac)	287.53	156.40	460.95	29.63	44.78	68.83	34.04	31.367	17.03	52.95	114.35	17.91
Runtime (ms)	78.02	49.86	112.71	54.18	48.82	34.35	11.35	132.03	22.62	53.85	9540.21	5.14

Table 5

Comparison of computational efficiency among representative dehazing models evaluated on 512 × 512 images.												
Method	FFA	SGID	C2PNet	Dehaze Former -B	SANet	DCMPNet	DEA-Ne t	MB-Taylor FormerV2- B	TUR	SGDN	<i>Learning Hazing to Dehazing</i>	Ours
Params (M)	4.46	13.87	7.17	2.52	3.81	17.36	3.65	2.63	2.60	11.08	1228.89	17.51
FLOPs (GMac)	1150.13	625.61	1843.82	93.99	188.08	275.08	136.17	125.47	68.01	211.86	448.63	71.64
Runtime (ms)	479.2	189.12	682.64	253.26	219.25	155.87	51.94	536.15	101.38	224.63	19870.79	19.35

4.6. Ablation Study

To evaluate the design choices of ZID-Net, we conduct ablation studies on the Real-World, StateHaze1k, and SOTS-Outdoor datasets. We first analyze the contributions of the main network components and compare ZI-PPH with other auxiliary supervision heads. We then examine the correction fields produced by these heads, compare different timestep sampling strategies, evaluate different high-frequency extraction operators, and study the effects of the training losses and their weights.

Effectiveness of Network Components. As shown in Table 6, the full ZID-Net achieves the best performance across all testing scenarios, with the corresponding visual results for the Real-World Thin dataset presented in Fig. 11. Removing the structural branch leads to substantial PSNR decreases of 2.61 dB on the Real-World Thin dataset and 2.18 dB on the Real-World Dense dataset. As illustrated in Fig. 11, this quantitative decline is accompanied by

Comparison of computational efficiency among representative dehazing models evaluated on 256 × 256 images.												
Method	FFA	SGID	C2PNe t	Dehaze Former- B	SAN et	DCMPNet	DEA-Ne t	MB-Taylor FormerV2-B	TUR	SGD N	<i>Learning Hazing to Dehazing</i>	Ours
Params (M)	4.46	13.87	7.17	2.52	3.81	17.36	3.65	2.63	2.60	11.08	1228.89	17.51
FLOPs (GMac)	287.53	156.40	460.95	29.63	44.78	68.83	34.04	31.367	17.03	52.95	114.35	17.91
Runtime (ms)	78.02	49.86	112.71	54.18	48.82	34.35	11.35	132.03	22.62	53.85	9540.21	5.14

Table 5

Comparison of computational efficiency among representative dehazing models evaluated on 512 × 512 images.												
方法	FFA	SGID	C2PNet	Dehaze Former -B	SANet	DCMPNet	DEA-Ne t	MB-Taylor FormerV2- B	TUR	SGDN	<i>Learning Hazing to Dehazing</i>	Ours
Params (M)	4.46	13.87	7.17	2.52	3.81	17.36	3.65	2.63	2.60	11.08	1228.89	17.51
FLOPs (GMac)	1150.13	625.61	1843.82	93.99	188.08	275.08	136.17	125.47	68.01	211.86	448.63	71.64
Runtime (ms)	479.2	189.12	682.64	253.26	219.25	155.87	51.94	536.15	101.38	224.63	19870.79	19.35

4.6. Ablation Study

为评估ZID-Net的设计选择，我们在Real-World、StateHaze1k和SOTS-Outdoor数据集上进行了消融研究。我们首先分析了主要网络组件的贡献，并将ZI-PPH与其他辅助监督头进行了比较。随后，我们检查了这些预测头产生的校正场，比较了不同的时间步采样策略，评估了不同的高频提取算子，并研究了训练损失及其权重的影响。

网络组件的有效性。如表6所示，完整的ZID-Net在所有测试场景中都取得了最佳性能，其在真实世界薄雾数据集上的对应可视化结果如图11所示。移除结构分支会导致峰值信噪比大幅下降——在真实世界薄雾数据集上下降2.61 dB，在真实世界浓雾数据集上下降2.18 dB。如图11所示，这种量化指标的下降伴随着

noticeable structural blurring, emphasizing the critical role of explicitly modeling high-frequency physical residuals. Similarly, discarding LGCBs or replacing DFAB with simple concatenation degrades metrics and reduces visual contrast across both datasets. Notably, training without the ZI-PPH branch reduces the PSNR from 21.83 dB to 19.89 dB under thin haze and from 20.87 dB to 19.27 dB under dense haze. These metric losses, alongside the residual haze visible in the ablation images in Fig. 11, demonstrate that the zero-inference diffusion prior provides essential structural regularization for the shared bottleneck.

Table 6

Ablation study on the architectural components of ZID-Net. The evaluation is conducted on the Real-World Thin and Real-World Dense datasets. Best results are highlighted in bold.

Method	Real-World Thin		Real-World Dense	
	PSNR↑	SSIM↑	PSNR↑	SSIM↑
w/o Structural Branch	19.22	0.658	18.69	0.495
w/o ZI-PPH	19.89	0.701	19.27	0.534
w/o LGCBs	20.91	0.727	19.86	0.576
w/o DFAB	20.10	0.741	19.52	0.579
Full	21.83	0.776	20.87	0.601



Fig. 11. Visual ablation results of ZID-Net on the Real-World Thin dataset.

Comparison of Auxiliary Supervision. We attach four alternative prediction heads to the shared bottleneck to compare different forms of auxiliary supervision. Following the atmospheric scattering model, the A -Head predicts global atmospheric light (A) as a three-dimensional color vector, while the t -Head estimates a single-channel transmission map (t). The joint $A + t$ Head combines both predictions. The Residual-Head instead directly reconstructs the three-channel haze residual. Table 7 compares the restoration performance of the resulting variants, among which ZI-PPH achieves the highest quantitative

明显的结构模糊，这凸显了显式建模高频物理残差的关键作用。同样地，舍弃LGCBs或用简单拼接替代DFAB也会导致两个数据集上的指标下降和视觉对比度降低。值得注意的是，不使用ZI-PPH分支进行训练时，峰值信噪比在薄雾条件下从21.83 dB降至19.89 dB，在浓雾条件下从20.87 dB降至19.27 dB。这些指标损失，加上图11消融图像中可见的残余雾霾，表明零推理扩散先验为共享瓶颈提供了至关重要的结构正则化。

Table 6

Ablation study on the architectural components of ZID-Net. The evaluation is conducted on the Real-World Thin and Real-World Dense datasets. Best results are highlighted in bold.

Method	Real-World Thin		Real-World Dense	
	PSNR↑	SSIM↑	峰值信噪比↑	结构相似性↑
w/o Structural Branch	19.22	0.658	18.69	0.495
w/o ZI-PPH	19.89	0.701	19.27	0.534
w/o LGCBs	20.91	0.727	19.86	0.576
w/o DFAB	20.10	0.741	19.52	0.579
Full	21.83	0.776	20.87	0.601



Fig. 11. Visual ablation results of ZID-Net on the Real-World Thin dataset.

辅助监督对比。我们将四个备选预测头接到共享瓶颈上，以比较不同形式的辅助监督。遵循大气散射模型， A -Head 将全局大气光 (A) 预测为一个三维颜色向量，而 t -Head 估计单通道透射图 (t)。联合 $A + t$ Head 结合了这两种预测。Residual-Head 则直接重建三通道雾霾残差。表 7 比较了所得各变体的恢复性能，其中 ZI-PPH 取得了最高的定量

results. To further investigate the reason for this performance difference, we conduct a correction-field analysis. As shown in Fig. 12, for each trained variant m , we derive its correction field as $\Delta I_m = \hat{I}_m - I_h$, where $\hat{I}_m$ denotes the restored output of the corresponding variant. The reference correction field is defined as $\Delta I_{\text{GT}} = I_c - I_h$. Consistent with the results in Table 7, ZI-PPH produces a correction field that is closest to the reference, achieving the highest spatial correlation and the lowest MAE. In comparison, the simpler auxiliary heads produce less spatially consistent correction patterns. These results suggest that diffusion-based noise-prediction supervision provides a more effective training signal for spatially varying haze than direct physical-parameter or residual regression. Since all auxiliary heads are removed during testing, these improvements introduce no additional inference cost.

Table 7

Quantitative comparison of different auxiliary supervision heads during training on the Real-World Thin dataset. All auxiliary heads are detached during inference. Best results are highlighted in bold.

Method	Real-World Thin		
	PSNR↑	SSIM↑	Training overhead (s/batch) ↓
Backbone only	19.89	0.701	1.02
+ A -Head	19.92	0.709	1.10
+ t -Head	19.71	0.711	1.15
+ A + t -Head	20.68	0.724	1.16
+ Residual Head	20.43	0.708	1.16
+ ZI-PPH (Ours)	21.83	0.776	1.21

结果。为了进一步探究这种性能差异的原因，我们进行了校正场分析。如图 12 所示，对于每个训练后的变体 m ，我们将其校正场推导为 $\Delta I_m = I_m - I_h$ ，其中 I_m 表示相应变体的复原输出。参考校正场定义为 $\Delta I_{\text{GT}} = I_c - I_h$ 。与表 7 的结果一致，ZI-PPH 产生的校正场最接近参考值，实现了最高的空间相关性和最低的 MAE。相比之下，较简单的辅助头产生的校正模式在空间上不够一致。这些结果表明，对于空间变化雾，基于扩散的噪声预测监督比直接进行物理参数回归或残差回归提供了更有效的训练信号。由于所有辅助头在测试时都被移除，这些改进不会带来额外的推理成本。



表 7不同辅助监督头在真实世界薄雾数据集上训练期间的定量比较。所有辅助头在推理期间均被剥离。最佳结果以粗体突出显示。

方法	真实世界薄雾		
	峰值信噪比↑	结构相似性↑	训练开销（秒/批次）↓
Backbone only	19.89	0.701	1.02
+ A -Head	19.92	0.709	1.10
+ t -Head	19.71	0.711	1.15
+ A + t -Head	20.68	0.724	1.16
+ Residual Head	20.43	0.708	1.16
+ ZI-PPH (Ours)	21.83	0.776	1.21

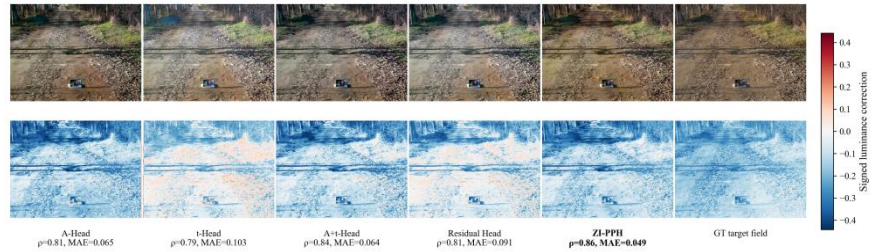


Fig. 12. Correction-field comparison of different auxiliary-supervision variants. ρ denotes the spatial correlation coefficient, and MAE denotes the mean absolute error; both are computed against the reference correction field ΔI_{GT} . Higher ρ and lower MAE indicate better agreement.

Effect of Timestep Sampling. We further compare the proposed adaptive strategy with uniform timestep sampling on StateHaze1k. As shown in Table 8, adaptive sampling improves all three haze levels, increasing the average PSNR by 1.25 dB and SSIM by 0.013. Unlike uniform sampling, it assigns smaller timestep ranges to light-haze samples to avoid excessive corruption of fine details, while exposing heavier-haze samples to a broader noise range to strengthen structural modeling. This better matches the perturbation strength to the actual degradation level and provides more effective supervision for the shared bottleneck.

Table 8

Effect of timestep sampling on StateHaze1k. The best result in each column is highlighted in bold.

Sampling strategy	Thin		Moderate		Thick	
	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
Uniform timestep sampling	24.40	0.920	25.95	0.936	21.76	0.849
Adaptive timestep sampling (Ours)	25.26	0.928	27.71	0.947	22.91	0.869

Analysis of Structural Prior Inputs. As detailed in Table 9, the proposed Color Laplacian achieves the highest PSNR and SSIM alongside the lowest ΔE_{ab} [42] and ΔE_{00} [43] color distortion metrics. This distinct advantage stems from preserving signed cross-channel variations, which alternative

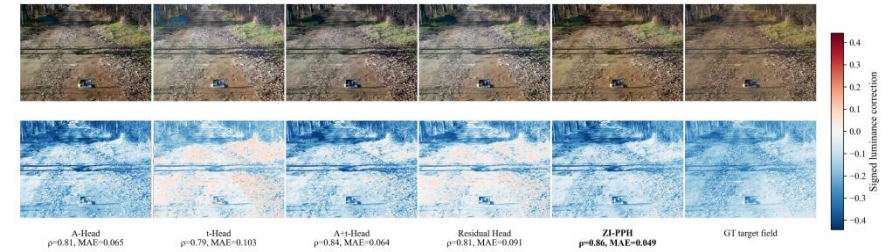


图 12. 不同辅助监督变体的校正场对比。 ρ 表示空间相关系数，MAE 表示平均绝对误差；二者均以参考校正场 ΔI_{GT} 为基准计算。 ρ 越高、MAE 越低，表示一致性越好。

时间步采样的影响。 我们进一步在 StateHaze1k 上将所提出的自适应策略与均匀时间步采样进行了比较。如表 8 所示，自适应采样在三个雾霾等级上均有所提升，平均 PSNR 提高 1.25 dB，SSIM 提高 0.013。与均匀采样不同，它为轻雾样本分配更小的时间步范围，以避免对细节造成过度破坏，同时让重雾样本暴露在更宽的噪声范围中，以强化结构建模。这使得扰动强度与实际退化水平更加匹配，并为共享瓶颈提供更有效的监督。

表 8 时间步采样在 StateHaze1k 上的影响。每列最佳结果以粗体标出。

采样策略	Thin		中等雾		Thick	
	峰值信噪比 $\uparrow$	结构相似性 $\uparrow$	峰值信噪比 $\uparrow$	结构相似性 $\uparrow$	峰值信噪比 $\uparrow$	结构相似性 $\uparrow$
均匀时间步采样	24.40	0.920	25.95	0.936	21.76	0.849
自适应时间步采样 (本文方法)	25.26	0.928	27.71	0.947	22.91	0.869

结构先验输入分析。 如表9所示，所提出的彩色拉普拉斯算子取得了最高的峰值信噪比和结构相似性，同时具有最低的 ΔE_{ab} [42]和 ΔE_{00} [43] 颜色失真指标。这一显著优势源于其保留了带符号的跨通道变化，而其他

extractors inherently destroy. Canny and the Gray Laplacian collapse the input into a single luminance channel, eliminating crucial cross-channel degradation differences. Operators such as Sobel, Prewitt, and Scharr process independent RGB channels. However, they compute non-negative gradient magnitudes and inherently discard critical directional signs. This physical limitation is also shared by Gabor filters, which compress high-frequency representations via absolute averaging. The Color Laplacian circumvents these issues by computing signed second-order residuals directly across the RGB channels, explicitly capturing both directional transition signs and color-specific attenuation priors. The practical impact of retaining these uncompressed structural and chromatic details is visually evident in Fig. 13.

Table 9

Quantitative comparison of different high-frequency operators for the structural prior. Best results are highlighted in bold.

Method	Real-World Thin				SOTS-Outdoor			
	PSNR↑	SSIM↑	ΔE_{ab} ↓	ΔE_{00} ↓	PSNR↑	SSIM↑	ΔE_{ab} ↓	ΔE_{00} ↓
Canny	18.17	0.634	14.025	11.199	34.53	0.993	2.093	1.557
Gabor	17.79	0.674	15.803	12.448	35.08	0.980	2.368	1.900
Sobel	17.59	0.613	14.562	11.666	35.88	0.986	1.847	1.406
Prewitt	17.58	0.613	14.571	11.674	35.88	0.986	1.848	1.407
Scharr	17.61	0.615	14.569	11.672	35.83	0.986	1.847	1.403
Gray Laplacian	21.74	0.761	10.446	8.476	38.25	0.992	1.516	1.143
Color Laplacian (Ours)	21.83	0.776	10.401	8.409	38.73	0.993	1.408	1.078

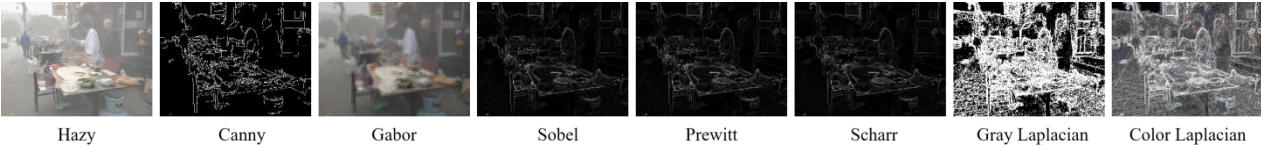


Fig. 13. Visual comparison of high-frequency features extracted by different operators.

Ablation of Loss Functions and Weights. Table 10 evaluates the

提取算子本质上会破坏这些变化。Canny 算子和灰度拉普拉斯算子将输入压缩为单一的亮度通道，消除了关键的跨通道退化差异。Sobel、Prewitt 和 Scharr 等算子处理独立的 RGB 通道。然而，它们计算的是非负梯度幅值，本质上丢弃了关键的方向符号。这一物理局限同样存在于 Gabor 滤波器中，后者通过绝对平均压缩高频表示。彩色拉普拉斯算子通过直接在 RGB 通道间计算带符号的二阶残差来规避这些问题，显式地同时捕获方向过渡符号和颜色特异性衰减先验。保留这些未压缩的结构与色彩细节的实际效果在图13中直观可见。

表 9针对结构先验的不同高频算子的定量比较。最佳结果以粗体高亮显示。

方法	真实世界薄雾				SOTS-Outdoor			
	峰值信噪比↑	结构相似性↑	ΔE_{ab} ↓	ΔE_{00} ↓	峰值信噪比↑	结构相似性↑	ΔE_{ab} ↓	ΔE_{00} ↓
Canny	18.17	0.634	14.025	11.199	34.53	0.993	2.093	1.557
Gabor	17.79	0.674	15.803	12.448	35.08	0.980	2.368	1.900
Sobel	17.59	0.613	14.562	11.666	35.88	0.986	1.847	1.406
Prewitt	17.58	0.613	14.571	11.674	35.88	0.986	1.848	1.407
Scharr	17.61	0.615	14.569	11.672	35.83	0.986	1.847	1.403
Gray Laplacian	21.74	0.761	10.446	8.476	38.25	0.992	1.516	1.143
Color Laplacian (Ours)	21.83	0.776	10.401	8.409	38.73	0.993	1.408	1.078

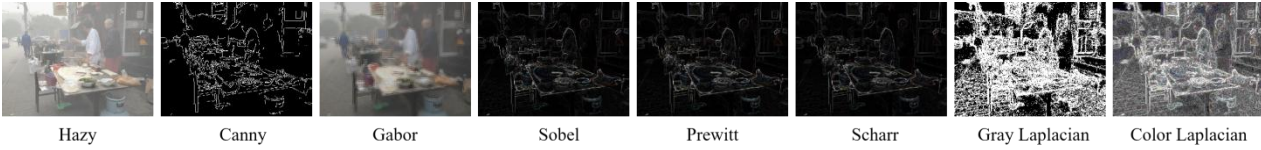


Fig. 13. Visual comparison of high-frequency features extracted by different operators.

Ablation of Loss Functions and Weights. Table 10 evaluates the

contributions and balance of the two auxiliary losses, with the reconstruction weight kept at 1.0. Removing the contrastive loss decreases PSNR by 1.12 dB and SSIM by 0.056, while removing the diffusion loss causes larger drops of 1.71 dB and 0.092, respectively. This indicates that diffusion supervision provides the stronger constraint for learning haze-related residual distributions, whereas contrastive supervision offers complementary feature discrimination. The weight analysis shows a non-monotonic effect: overly weak contrastive supervision cannot provide sufficient feature guidance, while an excessive weight competes with image reconstruction. Similarly, insufficient or excessive diffusion supervision disturbs the balance between residual modeling and restoration. Among the tested settings, the adopted weights provide the most balanced restoration performance.

Table 10
Ablation of loss terms and weights on the Real-World Dense dataset. λ_1 is fixed to 1.0. Best available values are highlighted in bold.

λ_2	λ_3	Real-World Dense		
		PSNR $\uparrow$	SSIM $\uparrow$	$\Delta \mathbf{E}_{00}\downarrow$
0.00	0.35	19.75	0.545	13.32
0.10	0.00	19.16	0.509	13.36
0.05	0.35	19.70	0.558	13.09
0.20	0.35	18.70	0.531	14.07
0.10	0.20	18.80	0.510	14.48
0.10	0.50	18.65	0.510	14.38
0.10	0.35	20.87	0.601	10.83

5. Conclusion

In this paper, we proposed ZID-Net, a novel frequency-spatial decoupled framework that addresses the conflict between the generative capability of diffusion models and the latency requirements of low-level vision tasks. During training, the Zero-Inference Prior Propagation Head introduces dynamic noise

贡献及两个辅助损失的平衡，重建权重保持为1.0。移除对比损失会使 PSNR下降1.12 dB、SSIM下降0.056，而移除扩散损失则会导致更大的下降，分别为1.71 dB和0.092。这表明扩散监督为学习雾霾相关残差分布提供了更强的约束，而对比监督则提供了互补的特征判别。权重分析显示出一种非单调效应：过弱的对比监督无法提供足够的特征引导，而过大的权重则会与图像重建竞争。同样，不足或过度的扩散监督会扰乱残差建模与图像复原之间的平衡。在测试的设置中，所采用的权重提供了最均衡的恢复性能。

表10在真实世界浓雾数据集上对损失项和权重的消融实验。 λ_1 固定为1.0。最佳可用值以粗体突出显示。

λ_2	λ_3	真实世界浓雾		
		峰值信噪比 $\uparrow$	结构相似性 $\uparrow$	$\Delta \mathbf{E}_{00}\downarrow$
0.00	0.35	19.75	0.545	13.32
0.10	0.00	19.16	0.509	13.36
0.05	0.35	19.70	0.558	13.09
0.20	0.35	18.70	0.531	14.07
0.10	0.20	18.80	0.510	14.48
0.10	0.50	18.65	0.510	14.38
0.10	0.35	20.87	0.601	10.83

5. 结论

本文提出了ZID-Net，一种新颖的频域-空间解耦框架，旨在解决扩散模型的生成能力与底层视觉任务延迟要求之间的冲突。在训练过程中，零推理先验传播头在不同时间步引入动态噪声预测，为共享瓶颈提供退化感知监督。

prediction across varying timesteps, providing degradation-aware supervision for the shared bottleneck. This diffusion branch is completely removed during inference, allowing the network to benefit from diffusion-based regularization without introducing additional computational overhead. To establish long-range spatial dependencies, Lightweight Global Context Blocks are incorporated into the backbone, while a parallel high-frequency branch employs the Channel-Spatial Laplacian Mask to suppress haze-amplified responses and preserve chromatic and structural details. These complementary representations are integrated through the Dynamic Feature Arbitration Block, which adaptively reweights semantic, structural, and upsampled features during image reconstruction. Extensive experiments on synthetic, real-world, and remote-sensing datasets demonstrate that ZID-Net achieves competitive restoration performance while maintaining deterministic and efficient inference. The current design does not explicitly adapt to unseen data distributions during inference, and its performance under substantial domain shifts, nighttime illumination changes, and near-complete scene occlusion remains to be further investigated. Future work will explore more general degradation representations and cross-domain restoration strategies to further improve robustness under diverse adverse conditions.

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该扩散分支在推理时被完全移除，使网络能够受益于基于扩散的正则化，而不会引入额外的计算开销。为了建立长程空间依赖，骨干网络中引入了轻量级全局上下文模块，同时并行的高频分支利用通道-空间拉普拉斯掩码来抑制雾霾放大响应，并保留色彩信息和结构细节。这些互补表示通过动态特征仲裁模块进行整合，该模块在图像重建过程中自适应地重新加权语义、结构和上采样特征。在合成、真实世界和遥感数据集上的大量实验表明，ZID-Net在保持确定性且高效推理的同时，实现了具有竞争力的恢复性能。当前设计在推理过程中未显式适应未知数据分布，其在显著域偏移、夜间光照变化和近乎完全场景遮挡条件下的性能仍有待进一步研究。未来工作将探索更通用的退化表示和跨域恢复策略，以进一步提高在多样化恶劣条件下的鲁棒性。

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